

Review Article

Advancements in Machine Learning-Enhanced Green Wireless Sensor Networks: A Comprehensive Survey on Energy Efficiency, Network Performance, and Future Directions

Kofi Sarpong Adu-Manu ¹, Emmanuel Amoako ¹, and Felicia Engmann ²

¹Department of Computer Science, University of Ghana, Accra, Ghana

²Department of Computer Science, Ghana Institute of Management and Public Administration, Accra, Ghana

Correspondence should be addressed to Kofi Sarpong Adu-Manu; ksadu-manu@ug.edu.gh

Received 21 May 2024; Accepted 2 April 2025

Academic Editor: Shonak Bansal

Copyright © 2025 Kofi Sarpong Adu-Manu et al. Journal of Sensors published by John Wiley & Sons Ltd. This is an open access article under the terms of the Creative Commons Attribution License, which permits use, distribution and reproduction in any medium, provided the original work is properly cited.

Wireless sensor networks (WSNs) are a collection of sensor nodes that collect data from the environment using wireless technology. WSNs have many applications in various domains, such as public utilities, industrial monitoring and control, and defense and military activities. However, WSNs have limited energy, short network lifetime, high bandwidth requirements, low throughput (TP), and unreliable connections. Green WSNs (GWSNs) are approaches that optimize energy consumption and enhance sustainable networks. Despite these advancements, nonadaptability to dynamic network conditions and the use of static historical data necessitates introducing machine learning (ML) techniques to address these challenges. GWSNs aim to reduce energy consumption and environmental impact, while ML techniques will improve data processing and network performance. This paper surveys recent advances in ML-based GWSNs, covering different aspects such as network structure, data exchange, location information, quality of service (QoS), and multiple path support. We also present the performance metrics, implementation issues, and future trends in ML-based GWSNs. The paper introduces a new taxonomy categorizing ML applications based on network architecture, data sharing, location data, multipath support, and QoS. The survey findings show that ML-based GWSNs can achieve up to 50% energy savings, 30% TP improvement, and a 40% delay reduction (DR) compared to conventional WSNs.

1. Introduction

Wireless sensor networks (WSNs) are collections of sensor nodes capable of sensing and transmitting sensed data to a remote monitoring center via wireless technology (e.g., wireless fidelity [WiFi], Worldwide Interoperability for Microwave Access [WiMAX], ZigBee, and Global System for Mobile Communication [GSM]). The WSN architecture allows for the simultaneous collection of data from spatially dispersed locations and is tailored to specific situations in deployable environments [1]. Energy is a critical factor in regulating growth and dictating many elements of human activity [2]. Globally, energy crises and carbon emissions have become significant concerns [3]. Hence, deploying a practically feasible solution that presents unique consumption and production processes to reduce environmental carbon emissions is

necessary. This environmentally friendly requirement for continuous communication necessitates using sustainable approaches for WSN implementation, that is, green WSN (GWSN). In this study, a network is considered green if it uses less energy and prolongs its lifetime. Hence, GWSNs combine techniques that support energy conservation, harvesting, and reduction [4, 5]. In Figure 1, a combination of the components of the GWSN is presented. Energy harvesting, conservation, and reduction techniques form the green energy framework for WSNs because they ensure that energy usage is reduced to the optimum minimum during network operation. While energy harvesting ensures a continuous energy supply to the network from ambient sources, energy conservation reduces the energy usage from the sensor node's components. Energy reduction mainly applies techniques that reduce

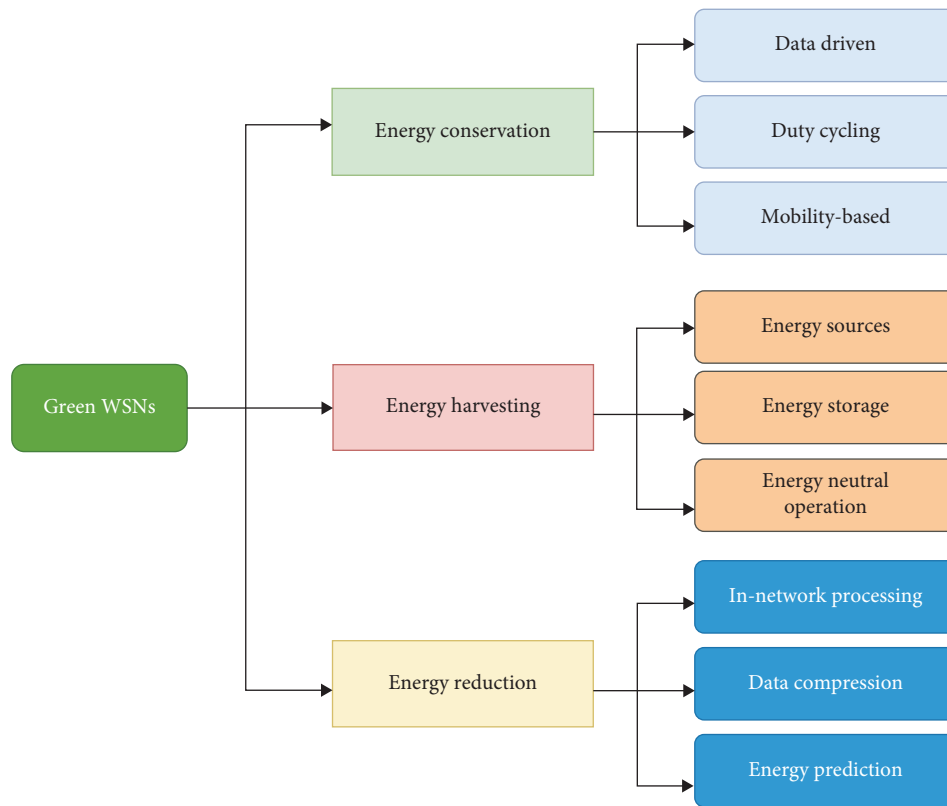


FIGURE 1: Components of the green energy framework.

the number of packets in the network during communication without compromising the quality of the monitoring process.

In GWSNs, energy conservation applies data-driven techniques, duty cycling, and mobility schemes, as indicated in Figure 2. Energy harvesting focuses on replenishing the energy stored in ambient sources, using efficient storage devices for harvested energy, and providing options for energy-neutral network operation (ENO). To enhance the performance and sustainability of WSNs, GWSNs are a promising solution that integrates energy harvesting, duty cycling, and data reduction to optimize energy consumption and prolong the network's lifetime. However, traditional techniques in WSN optimization often rely on static data and network configurations, which limits their ability to adapt to the dynamic network conditions of GWSNs [6]. Therefore, machine learning (ML) methodologies promise a transformative approach to incorporating intelligent, data-driven advantages to decision-making. ML enables predictive data analysis and autonomous decision-making by leveraging historical data and real-time monitoring of ambient data. ML could also identify and detect abnormal sensor readings and faults indicating faulty sensor readings or damage. They also optimize efficient bandwidth usage and computational resource management of constrained WSN resources [7].

With the growing demand for energy-efficient and environmentally sustainable WSNs, this study aims to survey ML protocols for GWSNs comprehensively. WSNs are crucial in various applications, such as environmental monitoring, healthcare, military, agriculture, and the Internet of Things

(IoT). However, the energy constraints of battery-powered sensor nodes and the lack of continuous green energy sources pose significant challenges for GWSNs. Efficient protocols, particularly medium access control (MAC) protocols, are essential for energy conservation and prolonging network lifetime by optimizing data transmission schedules.

This survey presents the current state-of-the-art ML protocols for GWSNs and the optimization techniques implemented in the literature. ML implementations are categorized based on network architecture, data sharing, location data, service quality, and multipath support. This novel taxonomy enables researchers and network experts to understand current trends, application-specific requirements, and challenges and facilitates continuous environmental monitoring using GWSNs.

The main contributions of this study include the following:

- Provides an in-depth analysis of current ML developments for GWSNs.
- Presents new taxonomy classifying ML-based GWSNs based on network architecture, data sharing, location data, service quality, and multipath support.
- Evaluate the performance metrics for evaluating ML-based GWSN efficiency.
- Identify and discuss the challenges of real-world ML-based GWSN deployment and propose potential solutions.
- Critically reviews a range of ML techniques used in GWSNs (data aggregation, MAC, clustering, routing).

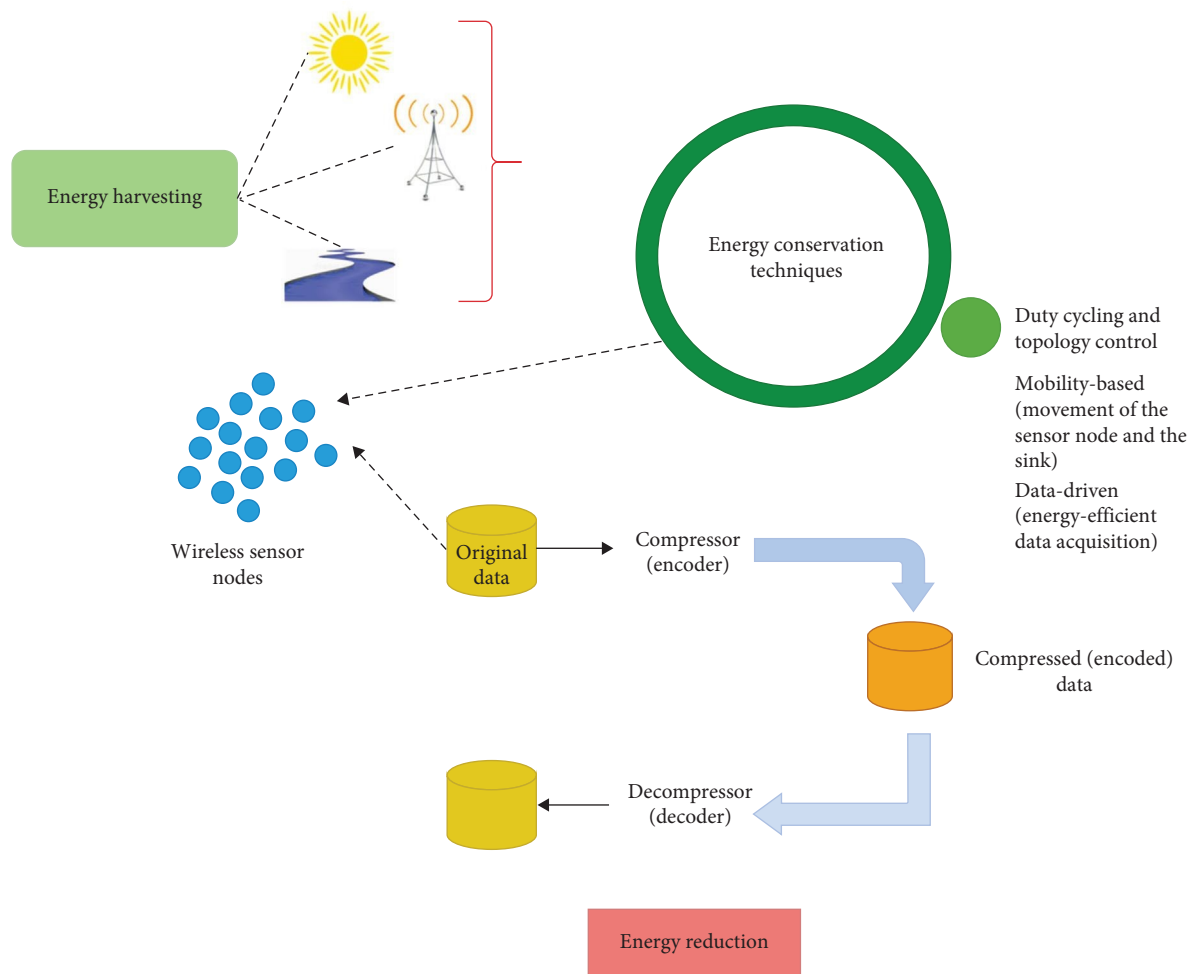


FIGURE 2: Implementations of green wireless sensor networks.

- Highlights adaptability of ML-driven GWSNs across some applications like agriculture, healthcare, military, and smart cities.

Despite these advances, several critical research gaps remain in ML-driven GWSNs. Challenges such as dependency on variable green energy sources, limited computational resources of sensor nodes, and adaptability of ML algorithms across diverse applications hinder the efficiency and sustainability of GWSNs. Furthermore, the lack of real-world deployment and validation underscores the need for practical assessment, whereas emerging ML approaches, such as federated learning, require further exploration. Security and privacy concerns in data-sensitive applications highlight the need for secure and privacy-preserving ML solutions. Addressing these gaps is essential in advancing GWSNs for broader and more sustainable applications.

The remainder of this paper is organized as follows: Section 2 describes the GWSN architecture and details the components of the sensor node. Section 3 discusses the role of ML in GWSNs. Section 4 presents the methodology of this study. Section 5 discusses energy-efficient protocols (routing, MAC) and highlights ML protocols and techniques for GWSNs.

Section 6 discusses protocol integration and performance metrics for GWSNs. In Section 7, we present the implementation of ML for GWSNs. Finally, Section 8 concludes the paper.

2. GWSN Architecture

A green network consists mainly of sensor nodes that conserve energy from the computing subsection of the sensor node and reduce the frequency of communication from the radio and antenna that form the communication subsection, as illustrated in Figure 3. The sensing subsection detects and measures ambient physical parameters such as temperature, pH, light, motion, and pressure to provide real-time monitoring of the environment. The computing subsection processes the data obtained, while the communication subsection transmits the data from sensor nodes to the base station [8]. Energy harvesting converts energy from ambient sources and either uses energy directly or is stored in energy receptacles for future use [5]. Sensor nodes are tiny devices that include sensors, transceivers, processing, and power units. Over the last two decades, using energy harvesting to power these sensor nodes has gained traction, resulting in the emergence of energy-harvesting WSNs (EH-WSNs) [5].

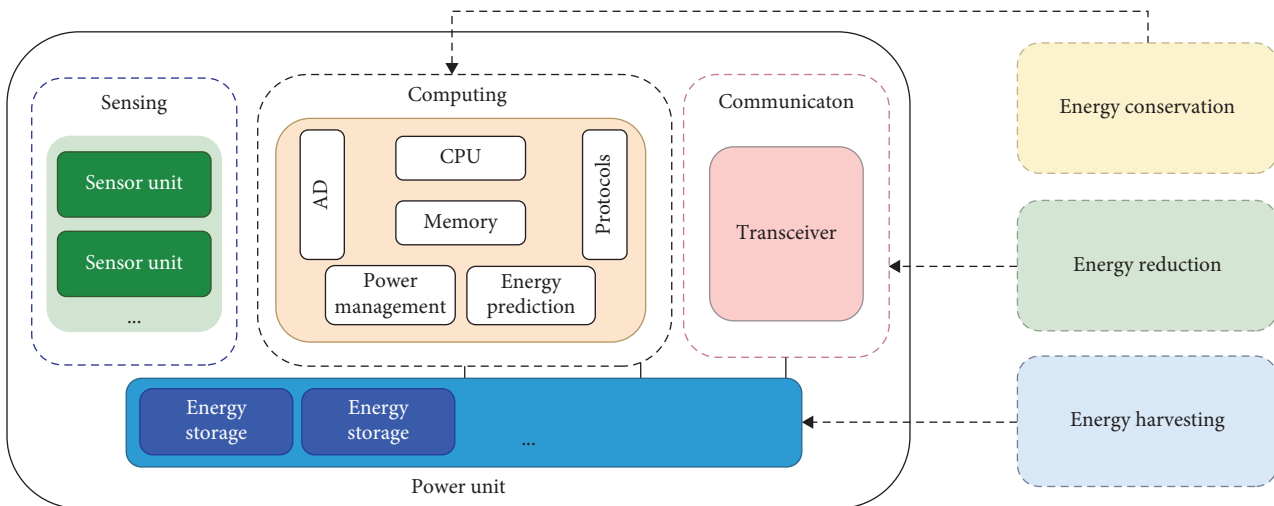


FIGURE 3: Green wireless sensor network (GWSN) sensor node architecture.

In the context of GWSNs, optimizing all sensor node components to minimize energy consumption and extend the network's operational lifetime and uptime is imperative. The energy-harvesting design alternatives [9] for GWSNs include the following:

1. Utilizing hybrid energy sources, such as solar, wind, and vibration, to provide continuous and reliable power to sensor nodes. ML approaches can be employed to optimize energy management and allocation.
2. Implementing ML techniques to adjust the beamforming vectors and reflecting coefficients of intelligent reflecting surfaces (IRSs). IRS can enhance wireless channel quality and reduce transmission power consumption.
3. Adopting cooperative communication and network coding to decrease sensor node transmissions and retransmissions. ML approaches can be used to select the optimal relay nodes and coding schemes.

These strategies represent promising avenues for future research aimed at enhancing the efficiency and sustainability of GWSNs.

In the literature, various energy-harvesting approaches include solar, radiation, vibration, and wind [5, 10, 11]. Solar energy, with a power density of 15 mW/cm^2 (outdoors on sunny days), is the most prevalent due to its environmental abundance [5]. Sah et al. [11] focused on solar energy for outdoor applications to enhance energy conversion efficiency, emphasizing green agriculture, green IoT, and solar streetlights. Their approach incorporated a short-term prediction model for outdoor solar energy harvesting. In [10], Adu-Manu et al., the energy harvesting approach integrated a pro-energy mechanism, utilizing a tuning factor and a fine adjustment index. Sarang et al. [12] simulated continuous energy harvesting over 96 consecutive hours under dynamic solar energy harvesting conditions. Their approach employed a self-adaptation technique to minimize energy consumption. When operating in a

green sensor network, amalgamating the methods discussed in this section may be used to optimize the node and network resources. These techniques support the network's bandwidth and available energy while reducing the end-to-end delay and frequent transmission loss. Consequently, the operation of a green network ensures that it remains active and efficient in performing its functions.

3. Role ML in GWSNs

ML learns data collection patterns over time and optimizes the process using algorithms and learning models [13]. ML algorithms can predict improvements as the dataset fed to the system grows and progresses, with researchers designing reliable and computationally viable techniques and algorithms [14]. These techniques and algorithms have been used extensively in bioinformatics, speech recognition, spam detection, computer vision, fraud detection, robotics, software engineering, genetic sequence analysis, compiler optimization, customer evaluation, and, recently, WSNs. Methods and algorithms are drawn from various disciplines, including statistics, mathematics, neuroscience, and computer science. ML is an essential technique for interpreting datasets the human brain cannot solve [15]. This paper discusses how ML can improve data processing and network performance in GWSNs and the challenges of applying ML in dynamic and resource-constrained environments. ML approaches used in solar energy harvesting include nonlinear autoregressive (NAR) neural networks [12], artificial neural network (ANN) models [16], and reinforcement learning models called SARSA learning working in an ENO [17]. ML also uses computational techniques to enhance machine performance by identifying and defining patterns and consistencies in data [14, 18]. ML is a valuable tool for the development of GWSNs. These networks aim to reduce energy consumption and use resources efficiently to minimize environmental impacts. For example, the European Union targets reducing greenhouse gas radiation by at least 50% by 2030 compared with 1990 [6]. WSNs can promote

energy efficiency (EE) in various human and economic applications. They can monitor pollution levels and traffic in smart cities and provide real-time information to encourage clean transportation options such as biking or walking [14, 19].

ML algorithms can help manage the operation of IoT nodes in smart cities where they are used for applications such as traffic monitoring, waste management, and building automation. However, there are challenges in designing these networks, including issues with coverage, connectivity, energy consumption, and bandwidth. They can potentially enhance the capabilities of GWSNs, making them more efficient and effective by making predictions that can aid in the reduction of continuous data collection while reducing energy consumption [14, 19, 20]. When the power of ML is applied to WSNs, the potential to enhance sensor network performance on specific tasks without reprogramming is guaranteed. ML is crucial for WSN applications for two reasons as follows:

1. WSNs gather new information regarding inaccessible hazardous sites (such as volcanic eruptions, wastewater, and water quality monitoring). WSN system designers can create ineffective solutions because of the unpredictability of the data stream. Relevant and efficient ML algorithms capable of adjusting the patterns of data streams in real-time are preferable for system designers [14, 18, 21].
2. The mobility of sensor nodes may cause changes in the network topology owing to dynamic changes in some WSN environments. For example, soil erosion or sea turbulence may cause a node's location to change due to drift in the water current, creating sensor networks that can adapt to their surroundings and function best in dynamic environments.

WSNs have been exposed to several research initiatives that have used advanced ML techniques to address the issues of the previous decade. Mittal et al. [22] suggested ML approaches for determining EE and detecting abnormalities in hybrid WSNs. Using neural networks, they designed two energy-efficient protocols (i.e., low-energy adaptive clustering hierarchy [LEACH] and energy-efficient sensor routing). In 2021, Pundir and Sandhu authored a publication using ML to improve the EE of WSN by reducing energy consumption and harvesting energy. ML techniques have been found to aid in reducing energy consumption and predicting the harvested energy from ambient and external sources [23].

ANNs, fuzzy logic maps, and support vector machines are used to monitor the fitness of electric vehicles. It also predicts the remaining life of a permanent magnet in the permanent magnet synchronous motor [24]. The prediction-based adaptive duty cycle MAC protocol (PADC_MAC) uses current and future harvested energy information using NAR neural networks to achieve better prediction accuracy in dynamic energy-harvesting scenarios [12].

In solar energy harvesting, where the energy source is not constant, and its intensity varies during the day, predicting the future energy to be harvested enables the development of efficient communication protocols [10]. Because ML algorithms

are designed to improve EE and increase the network lifetime of sensor nodes in WSNs, they have become indispensable in today's GWSN environment [25, 26]. ML techniques focus on route-selection approaches to improve communication and performance. ML algorithms can accurately predict the energy consumption of nodes within sensor networks and enhance the quality of service (QoS) parameters related to reliability and other factors in GWSNs.

4. Method

This systematic review and analysis used the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA). Figure 4 summarizes the PRISMA-based paper selection approach. Four essential steps of the PRISMA technique—Identification, Screening, Verifying eligibility, and Selection—were applied in this study. Relevant articles required for the review were determined during the identification phase.

Since this study is a systematic review, it does not involve direct data collection or preprocessing. However, data preprocessing plays an essential role in empirical studies within ML in WSNs by improving data quality, enhancing model performance, and ensuring consistency in analysis. Some of the most used preprocessing techniques include data cleaning (handling missing values and removing inconsistencies), normalization and standardization (scaling numerical data for uniformity), feature engineering (creating or transforming variables), and data transformation (converting categorical variables into numerical representations). Outlier detection and dimensionality reduction methods are often employed to refine datasets for better analysis.

Different scientific databases and libraries were searched to identify and select papers for review, including Springer, ACM, Scopus, Web of Science, IEEE Xplore, and ScienceDirect. The search strings listed in Table 1 were used to search for papers across the selected databases and libraries.

The PRISMA approach identification process with the search strings in Table 1 yielded over 7000 documents. In this phase, we narrowed it to publications between 2018 and 2024 to highlight recent trends in ML implementation for green sensor networks.

The initial screening of the papers was performed based on their titles, and further screening was performed using the abstract, main text, sub-headings, conclusion, and language. Using these parameters, the number of papers was reduced to 240 from 938. During the eligibility check, the 240 papers were fully read based on the following criteria: (1) the papers should be either a journal or a conference, (2) the paper should deal with GWSNs and ML, (3) the paper should focus on energy-efficient approaches within the ML and GWSNs, and (4) the paper should present a clear methodology and results.

4.1. Qualitative Synthesis. This section presents a qualitative synthesis of the selected papers. This section focuses on the yearly distribution of publications between 2018 and 2024, the type of publication, venues where these papers were published, and distribution based on ML and GWSNs.

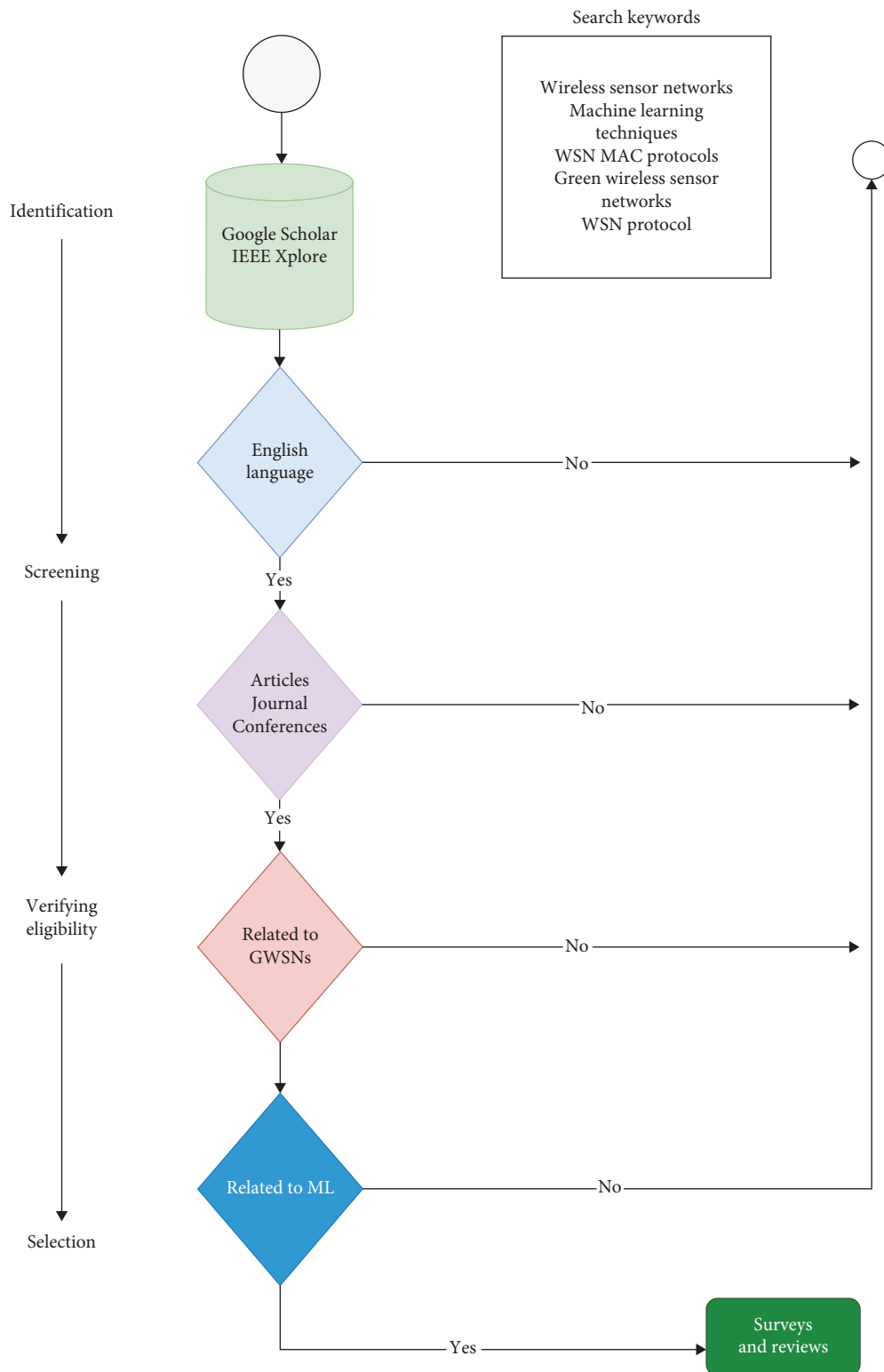


FIGURE 4: Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) flow diagram for paper selection.

4.1.1. Publications by Yearly Distribution. The yearly distribution of papers provides readers with recent developments in GWSN and ML. These papers were identified according to the *Publication Title*. Publications from 2018 to 2024 were considered for the evaluation and analysis. Google Scholar returned close to

428,000 search results when GWSNs and ML were used as search strings (keys). This demonstrates researchers' widespread interest in WSNs, especially in adopting newer ML techniques.

In this study, our interest was mainly related to how GWSNs and ML have been combined to address the energy

TABLE 1: Search strings used in search engines of academic literature databases.

Academic library	Search strings
IEEE-Xplore	1. (((“All Metadata”: machine learning)//filters applied: 2016–2020) 2. (((“All Metadata”: machine learning) OR “All Metadata”: wireless sensor networks)//filters applied 2010–2015) 3. (((“All Metadata”: machine learning) AND “All Metadata”: wireless sensor networks)//filters applied 2016–2020) 4. (((“All Metadata”: machine learning) AND “All Metadata”: wireless sensor networks AND “All Metadata”: MAC protocols)//filters applied 2016–2020) 5. (((“All Metadata”: machine learning) AND “All Metadata”: wireless sensor networks AND “All Metadata”: routing protocols)//filters applied 2016–2020)
Science Direct	1. Wireless sensor networks 2. Wireless sensor network protocols 3. Internet of Things 4. Machine learning techniques in wireless sensor network
Springer	1. Machine learning for WSN 2. Machine learning for WSN: review 3. Machine learning for green WSN: review 4. Machine learning for WSN: survey 5. Machine learning for green WSN: survey
ACM	1. Machine learning 2. Wireless sensor network 3. Machine learning for water quality 4. Machine learning for animal tracking applications 5. Machine learning for green wireless sensor networks
Scopus	1. ALL (“Computer science”) TITLE-ABS-KEY (“Machine learning protocols”) AND PUBYEAR = 2023
Web of Science	1. Energy-efficient protocols 2. Energy-efficient MAC protocols 3. Energy-efficient physical layer 4. Energy-efficient physical protocol 5. Energy-efficient MAC layer

challenge in sensor networks. In total, 240 papers were systematically reviewed to ascertain the usefulness of ML in GWSNs. Of the 240 papers, 56 (23.3%) matched our requirements at the prescreening stage.

ML models, particularly those used in classification and identification applications, provide accurate data records in WSNs [27]. The benefit of ML in GWSN systems is that it can significantly reduce the energy consumed by sensor nodes to improve network performance. Figure 5 presents the yearly distribution of ML-related publications in GWSNs over 7 years. The papers selected for the study on ML and GWSN were published between 2018 and 2019, accounting for ~50% of the total, as shown in Figure 5. We surveyed various papers from 2018 to 2024 on ML protocols for GWSNs, energy-efficient protocols at the physical, MAC, and network layers, QoS in GWSNs using ML approaches, and applications of ML in environmental monitoring applications. The distribution of the papers is presented in Table 2.

4.1.2. Distribution of Publication by Type. In this paper, we selected conference and journal papers published during the stated period for review. Other publication types, such as posters, abstracts, and patents, were eliminated during the screening process. Figure 6 shows the distribution percentages based

on publication type (i.e., conference or journal). Overall, 81% of the papers reviewed were published in journals, and the remaining 19% were published in conferences.

4.1.3. Distribution of Publications by Venue. Figure 7 presents the distribution of the selected papers according to publication venue. The articles were published in IEEE Xplore (25%), Science Direct (10%), Springer (4%), ACM (17%), Scopus (24%), and the Web of Science (20%).

In Figure 7, the distribution of the papers obtained from the various databases by the different publishers is presented, with 25% coming from IEEE-Xplore and 4% from Springer. Few studies have been conducted on GWSNs, and ML was obtained from the Scopus and Springer databases.

This study also investigated the distribution based on ML and GWSNs. After reviewing, organizing, and categorizing the GWSNs and ML research papers, we obtained the distribution presented in Figure 8.

5. Energy-Efficient Protocols for WSNs

This paper presents and evaluates the current developments in energy-efficient routing and MAC and ML protocols in WSNs. This section selected publications from 2018 to 2024 on routing and MAC layers and ML protocols, which focus on improving

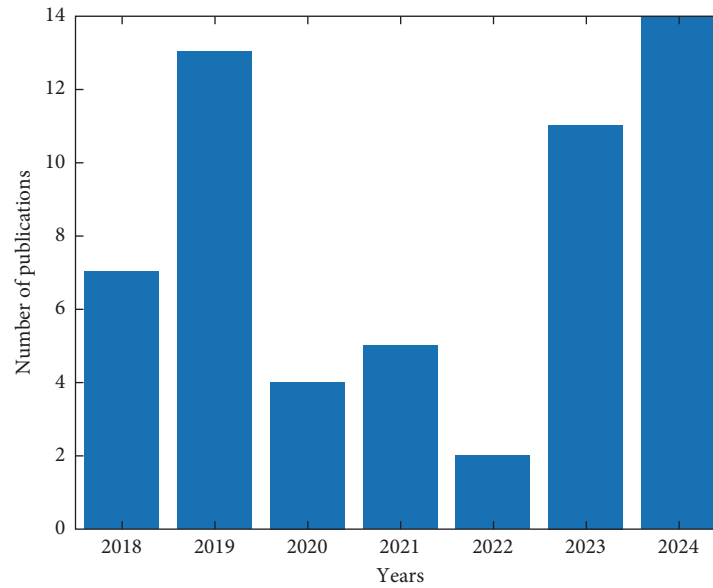


FIGURE 5: Publications by yearly distribution.

TABLE 2: Number of publications by classification areas.

Classification areas	2018	2019	2020	2021	2022	2023	2024
Machine learning protocols for green WSNs	1	1	3	2	0	0	0
Energy-efficient routing protocols	5	5	3	0	0	1	1
Energy-efficient MAC layer protocols	7	3	2	4	1	0	0
Energy-efficient physical layer protocols	2	3	2	0	0	0	0
Quality of service using machine learning	4	0	1	2	0	5	4
Machine learning for WSNs or green WSNs review or surveys	7	5	4	4	0	0	0
Machine learning for agricultural applications	2	2	4	3	3	8	6
Machine learning for water quality monitoring applications	4	5	4	3	4	4	6
Machine learning for animal tracking application	2	0	6	1	2	1	1
Machine learning for air pollution application	6	3	3	1	4	3	2
Machine learning for underground mining application	0	0	0	1	1	0	0
Machine learning-based energy-efficient approaches	0	2	0	0	2	0	0

EE and prolonging network lifetime, a critical challenge in WSNs. Sections 5.1, 5.2, and 5.3 present the overview, key findings, and limitations of enhanced routing protocols, improved MAC protocols, and energy-efficient ML protocols for WSNs.

5.1. Energy-Efficient Routing Protocols. This section presents and discusses five energy-efficient routing protocols designed to prolong network lifetime in WSNs. The protocols include multihop graph-based energy-efficient routing (MH-GEER), energy-efficient region source routing (ER-SR), Terrain-based Routing using fuzzy rules, cluster-based energy-efficient routing (CBE2R), and energy-efficient multistate routing systems (EE-MRP). We provide the following narrative summaries of these papers.

Rhim et al. [28] proposed an MH-GEER solution that uses a multihop graph-based approach for energy-efficient routing in

WSNs. MH-GEER aims to distribute energy usage evenly among clusters to prolong the network's lifetime. Given advances in wireless communication and miniature sensor nodes, sensor energy resources remain constrained, which is critical for establishing communication protocols. MH-GEER balances the energy consumption between clusters to extend the network lifetime. This approach has two phases: intercluster and clustering multihop selection (such as the operation of LEACH). Clustering is centralized, whereas cluster head selection is distributed. The algorithm uses probability to explore network energy levels and selects cluster head routes based on residual energy, node position, and traffic load. The clustering phase is based on the centralized creation of clusters and the distributed election of cluster heads, such as the LEACH. The routing phase establishes a dynamic multihop route between cluster heads and the base station. The approach examined the energy levels across the entire network and probabilistically used

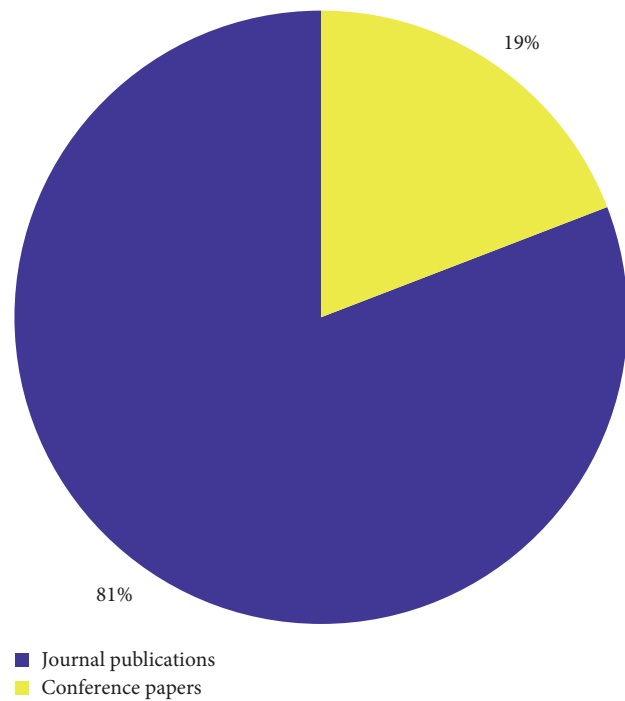


FIGURE 6: Publication distribution by type (conferences and journal papers).

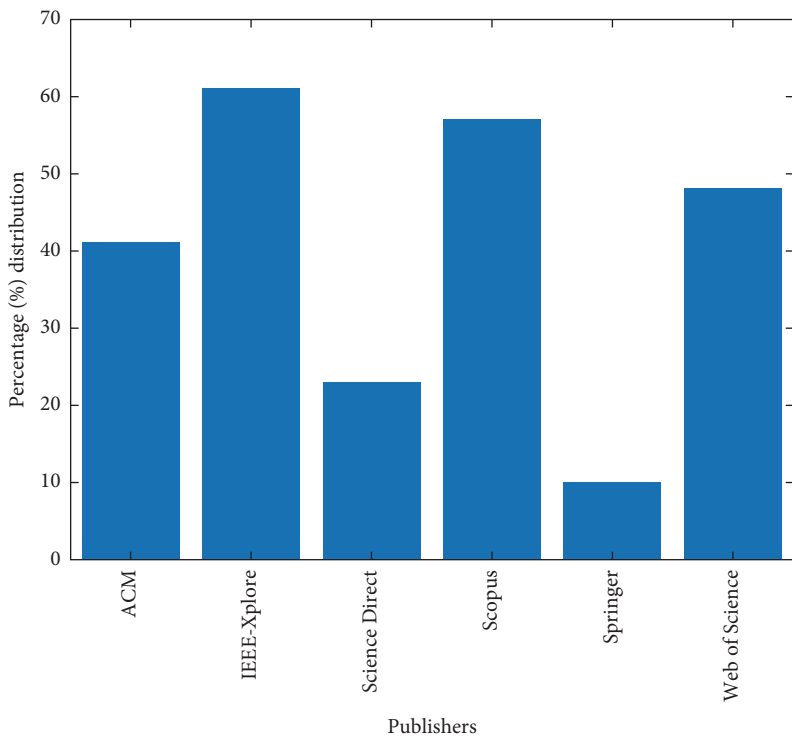


FIGURE 7: Distribution of publications by publishers.

this information to select the next hop. The proposed approach was compared with LEACH, and more energy was retained.

Xu et al. [29] presented a new ER-SR protocol to extend the network lifetime of WSNs. Sensor nodes, part of the IoT sensor

layer, are densely deployed in a hostile environment to monitor and perceive physical space changes. Traditional routing techniques frequently propagate across the entire network to find reliable paths, or they use cluster heads for data transfer, which

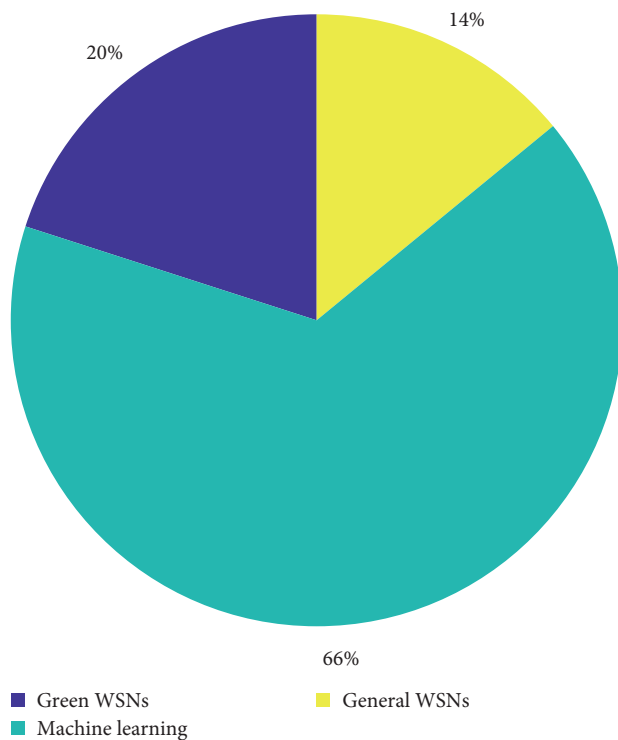


FIGURE 8: Distribution based on machine learning (ML), green wireless sensor networks (GWSNs), and wireless sensor networks (WSNs).

can be energy-intensive. The ER-SR method employs a distributed energy algorithm to pick source routing nodes dynamically and then computes the shortest routing paths for all nodes to balance the network's energy. In addition, an ant colony optimization method was used for each node to find the global optimal energy-efficient transmission line. Performance was evaluated based on the packet delivery ratio, network lifetime, and delivery delay. The ER-SR protocol uses a distributed energy region algorithm to choose high-residual-energy nodes as source-routing nodes dynamically. These source routing nodes determine the best source routing paths for all nodes, resulting in a balanced energy usage. An effective distance-based colony optimization technique is utilized to find the most energy-efficient pathways worldwide to minimize the transmission energy. The algorithm chooses high-energy nodes as source nodes based on signal intensity and data separation. It then uses a ring domain and a single-hop distance to determine the ideal transmission paths.

Pandiyaraju et al. [30] presented a new intelligent routing protocol proposed to improve the network lifetime and provide EE in the routing process, which provides data to the irrigation system. Many agricultural activities can be enhanced using sensor networks and data mining techniques. One of these activities is to regulate the quantity of water in cultivated fields. In recent years, WSNs have become an emerging technology in precision agriculture. The critical issue in the design of WSNs is the utilization of energy and the enhancement of the lifetime of the sensor nodes. This novel intelligent energy-efficient routing protocol uses fuzzy rules, and the protocol is called Terrain-based

Routing, which uses fuzzy rules for precision agriculture. The fuzzy inference system developed in this study was used to make the routing decisions. The system was implemented and compared with two routing algorithms: region-based routing and equalized cluster head election routing protocol.

Ahmed et al. [31] proposed the clustered-based energy-efficient routing (CBE2R) protocol for underwater WSNs (UWSNs), which addresses the challenges of designing efficient communication protocols in the underwater environment, such as propagation delay, high bit error rates, limited bandwidth, uncontrolled node mobility, water current, 3D deployment, and limited resources. The highest weighted value determines data forwarding in the clustered routing algorithm. The system uses courier nodes from the seafloor to the bottom layer to receive information from ordinary nodes and deliver it to the surface sink nodes at maximum power levels via courier nodes in different strata. The protocol uses high weights to select neighboring nodes from the selected couriers to the source nodes, with the chosen nodes no more than four hops away from the courier. The chosen courier nodes collect data packets from the source nodes via ordinary nodes and route them to sink nodes with the highest power levels installed on the water surface.

Khan et al. [32] proposed an energy-efficient routing protocol for WSNs. WSNs have recently attracted significant industrial and academic research attention. The primary reasons for recent research in this area include interest in applications such as military operations, surveillance systems, environmental event monitoring, healthcare, and human safety. This study proposes an energy-efficient multistage hierarchical routing protocol that uses forwarder nodes to reduce the energy consumed during communication. This approach uses a static clustering method to reduce the energy required for cluster head rotation. The authors described an EE-MRP, a WSN routing system. The main features of the EE-MRP protocol are as follows: it employs a dependable cluster head selection technique and static clustering to efficiently select cluster heads, thereby minimizing the need for frequent reclustering. The routing protocol includes a replacement mechanism to prevent the need for cluster head rotation. It uses forwarder nodes, where a node between the base station and cluster head collects the data and sends only the aggregated data to the cluster head. This minimizes the communication distance between the base station and the cluster head, resulting in lower energy consumption.

In Table 3, we present the limitations of some of the current developments in energy-efficient routing protocols in WSNs.

5.2. Energy-Efficient MAC Protocols. This section presents and discusses four energy-efficient MAC protocols designed to improve energy savings at the MAC layer in sensor networks. These include the energy-harvesting MAC (EH-MAC) protocol, dynamic channel coding MAC (DCC-MAC) protocol, quorum-based MAC protocol, and PSE²-MAC protocol. We provide the following narrative summaries of these papers.

Mehmood et al. [33] focused on the techniques of EE developed in the MAC layer. The proposed solution relies on energy sources, such as batteries and energy harvesting devices

TABLE 3: Limitations of energy-efficient routing protocol in WSNs.

Author(s)	Limitations
[28]	The authors did not implement the proposed protocol to evaluate its performance on large-scale networks. The proposed solution was compared with LEACH. Various new variants of LEACH have been proposed over the years, which could have resulted in better results. The network size was not varied enough to observe the impact of network size on MH-GEER
[29]	The theoretical analysis of network lifetime maximization, including all networks, overloads the routing protocol. Some routing protocols being compared were only mentioned in the short form, without their full meaning or adequate referencing. The types of nodes used were not adequately described, for example, homogeneous or not. The definition of energy consumption does not allow the energy of the individual nodes needed for the routing protocol to be performed
[30]	The authors did not consider making effective decisions regarding the terrain network
[31]	When the water pressure increases, the data delivery ratio of the multipath route is negatively affected, which does not make the approach robust for underwater propagation
[32]	For a routing protocol that seeks to portray energy efficiency, it would be prudent to include the energy consumption of the nodes/network as a parameter for network performance

(EHDs), to collect energy from the environment and analytic information, such as estimating the network lifetime, to achieve a potentially self-sustainable energy-efficient system. Because of their versatility, WSNs are being more widely used across various applications, even though energy consumption is a significant constraint for these networks. Therefore, it is critical to include constant and dependable energy sources. Because it entirely relies on unattached sources, such as batteries and EHDs, the MAC protocols created for WSN emphasize energy-saving solutions over other characteristics (throughput [TP], latency, jitter, etc.). This study used the D-cycle technique to describe the architecture of MAC protocols and divided them into synchronous and asynchronous categories. It examines the methods used by the current protocols to calculate the battery degradation and state of charge. Emphasizing the need for battery status data in MAC protocols, the study assesses current WSN MAC protocols and recommends new designs that include EHDs, duty cycle MAC, and battery state data to increase network lifetime.

Chehri [34] proposed a modified DCC-MAC protocol by introducing node energy consumption and switching the nodes' power levels simultaneously. DCC-MAC is designed for low-radiating-power ultra-wideband (UWB) ad hoc networks, combining MAC and PHY layers. The proposed model provides an energy-efficient protocol for e-health IoTs by extending battery lifetime using the energy levels of the nodes. The modified DCC-MAC combines adaptive pulse-position data modulation (PAM) modulation with switching power levels to extend its lifetime. The approach considers the exclusion region where nodes cannot transmit, whereas nodes outside this region can communicate simultaneously. The authors used a joint consideration of physical and multiple access layers to minimize the network's energy consumption and increase its lifetime. The proposed solution allows the interfering nodes to transmit only when they are outside the exclusion region of the destination node. This interference is mitigated using PAM, which produces a random process that describes the transmitted signal. The aim is to create a direct transmission link between the source and destination nodes such that the

transmitting nodes are not in the same interference region as the receivers.

Liu et al. [35] presented the quorum-based MAC protocol, which schedules separate and adaptive wake-up periods for nodes, decreasing idle listening and collisions while increasing network performance and extending network lifetime. They demonstrated a quantum time-slot adaptive condensing (QTSAC)-based MAC protocol for WSNs that achieves delay minimization and EE. Globally, energy crises and carbon emissions have become significant concerns, necessitating the creation of an energy-efficient MAC protocol, as millions of sensors used to monitor smart grids consume significant energy. The QTSAC allocates more quorum time slots (QTSs) for transmission to nodes further from the sink, assigning QTS only when the data are communicated. It addresses the issues raised by existing quorum-based MAC protocols, ensuring that only nodes with the same QTS can interact and provide extra QTS to the periphery nodes to reduce latency. Designing a new quorum system based on the MAC protocol includes maximizing the network lifetime, maximizing the effective utilization of energy, and minimizing the delay. Condensation of the QTS allows nodes to communicate with other nodes with the same QTS. This reduces the network latencies and collisions, leading to data loss. Peripheral nodes in the network have their QTSs increased to allow them to communicate in their active periods, thereby reducing latencies and balancing the energy in the network. Significant data production rates also indicate larger packet sizes and increased network delays.

Uddin et al. [36] proposed a predictive schedule-based energy-efficient MAC (PSE²-MAC) for asynchronous WSNs. The authors highlighted the beacon collision problem and proposed a possible solution with promising results. The use and scope of WSN have been further expanded with IoT and 5G networks. This paper highlighted the synchronous and asynchronous MAC protocols of WSNs, pointing out some challenges in their implementation, such as beacon interval nodes in an asynchronous protocol. They identified the challenges of both the sender-initiated and receiver-initiated protocols. It was noted that because nodes wake up independently of each

TABLE 4: Limitations of energy-efficient MAC protocol in WSNs.

Author(s)	Limitations
[28]	The approach to obtaining battery state information is not trivial. Additional devices may be required without considering the energy consumption and communication of these devices
[29]	It is a prerequisite that the complexity of implementing a channel encoder is not too large. The authors failed to pinpoint their selected approach for solving the problem exclusively. They suggest more than one approach used in the literature and assume that readers can choose the “obvious” solution. This makes the readability difficult
[30]	The network was limited to 100×100 , with less regard to more extensive networks. Future work will focus on the scalability of the proposed solution
[31]	The protocol can be tested on a large scale using modern simulation tools. This can be further improved when multiple senders send data to a single receiver using numerous channels. The simulator is not standardized and may introduce errors that cannot be verified. The network size was not varied enough in the simulation to note that changes in network size might affect the proposed solution

TABLE 5: Comparison of MAC protocol approaches for energy efficiency in WSNs.

Characteristics/authors	[33]	[37]	[35]	[36]
Energy sources and protocol classification	Utilizes batteries and EHDs; categorizes MAC protocols based on duty cycle, emphasizing asynchronous protocols	Dynamic adjustment based on nodes' energy levels, combining physical and MAC layer considerations	Quorum-based MAC with time-slot allocation for asynchronous wake-ups, suited for energy-critical scenarios	Focuses on asynchronous WSNs, with predictive wake-up scheduling tailored to beacon-based protocols
Communication strategy	Duty cycle-based transmission, using analytic battery state information for energy management	Uses adaptive power modulation with exclusion regions to control transmission	Quorum slots are allocated based on node distance to the sink, reducing collisions and idle listening	Predictive wake-up based on node activity, reducing idle listening in sender-initiated protocols
Interference handling	Minimal interference management due to emphasis on energy harvesting	Exclusion regions and PAM modulation control interference between source and destination nodes	Assigns extra quorum slots to peripheral nodes, reducing idle listening and collision at network edges	Beacon collision is mitigated by shifting wake-up cycles to avoid repeated collisions
Network scalability	Limited scalability due to reliance on additional devices for battery monitoring and energy management	Suitable for small-scale UWB networks but lacks clear guidance on handling complex, more extensive networks	Limited testing on small networks; scalability requires further testing on networks larger than 100×100	Tested on limited network size; scalability can improve with advanced simulations and multiple-channel use

other to sniff the network, a predictive wake-up technique is needed to reduce the frequency of collisions. The authors proposed a predictive wake-up technique for sender-initiated protocols to achieve improved energy savings. This approach predicts the receiver's wake-up to efficiently reduce the sender's idle listening period. In this approach, a neighbor wake-up table (NWT) is formed when a new node joins the network after a clear channel assessment (CCA). The node remains active with the necessary overhead costs during node discovery and NWT. When a beacon is transmitted and encounters a collision, the node shifts forward by a period known as the operational cost (SoC) start point to avoid collision in the next cycle. The proposed solution was compared with the traditional IEEE 802.15.4, using performance measures such as the number of collisions, control packet overhead, and TP.

In Table 4, we present the limitations of some of the current developments in energy-efficient MAC protocols in WSNs. Table 5 compares the MAC protocol approaches for the EE discussed in Section 5.2.

5.3. Efficient ML Protocols. This section presents and discusses four energy-efficient ML protocols proposed to optimize energy consumption in WSNs. These include ML-based data reduction algorithm (MLDR), whale optimization-driven criterion correlation (WOD-MCC) algorithms, ML-based optimization methods based on simulated annealing (SA), and scalable clustering algorithm and data aggregation (SCADA) using ANN. We provide the following narrative summaries of these papers.

Slim and Mitton [38] proposed an MLDR implemented at the sensor node level to optimize energy consumption and

TABLE 6: Limitations of energy-efficient ML protocol in WSNs.

Author(s)	Limitations
[28]	In their evaluation, the data reduction metrics of the temperature data. However, other important metrics, such as network latency, energy consumption, and scalability, were not considered. The study does not provide sufficient details on specific ML algorithms and their implementation in more complex agricultural environments
[29]	The computational complexity of ANN in WSN has not been evaluated or measured in simulations. Balancing ANN's memory and time complexities of an ANN in resource-constrained WSNs requires attention and optimality. In addition, the scalability of network deployment that supports the security and energy constraints of WSN with ANN requires attention not discussed in this paper
[30]	The network topology was not considered, even though some topologies and frequent changes in the topology might limit the approach's feasibility in real-world applications. Because labeled data are considered in the supervised approach, they are unsuitable for other applications without labeled data. Simulated annealing optimization has search space constraints that may lead to fewer optimal solutions with slower convergence speeds
[31]	The authors mentioned the implementation of the protocol in mobile environments, but the impact of mobility on data aggregation and clustering was not considered. The complexity and computational overhead of machine-learning algorithms have not been thoroughly analyzed. The results were obtained through simulations without considering its performance in the real world

bandwidth usage. Their implementation was conducted in an agricultural environment using real temperature data from the soil. ML trends in agriculture were reviewed, including predictive modeling, anomaly detection, data filtering and compression, adaptive learning, and EE. In agricultural WSNs, ML practices improve data reduction and EE and enhance decision-making capabilities for efficient and sustainable farming practices. The authors implemented a data reduction technique that detects abnormal situations in the meteorological data of the agricultural environment that may potentially harm farm produce. The authors compared the performance of MLDRs (i.e., hold [H] and buffer [B] scenarios). They concluded that MLDR-B computes newer trends in data to ensure data accuracy, whereas MLDR-H waits for new trends before transmission. They examined the use of ML to optimize data transmission and reduce energy consumption. Their results were validated using MATLAB simulations, and the number of transmissions using their two algorithms decreased from 192 to 25 (i.e., MLDR-H) and 27 (i.e., MLDR-B). However, from their analysis, MLDR-B preserves data integrity better than MLDR-H, but the latter is more energy-efficient.

Kang et al. [39] propose an optimization method based on SA. The authors employed a data reduction approach that removes redundant IoT sensors by correlating the sensed data from a smart home. The authors proposed an ML algorithm for an intelligent service model. The model is more intelligent than traditional systems because it optimizes the sensor node by adjusting sampling rates and transmission intervals. The approach also integrates ML techniques with meta-heuristics to suggest a new energy-saving strategy based on sensor data information. In addition, this approach uses the predictive performance of classifiers and information theory to identify the minimum subsets of sensors. This study employs a data-driven adaptive energy-saving technique using ML to optimize the sensor node operation of the device environment. The approach utilizes a hybrid filter-wrapper feature selection that identifies subsets of energy-saving nodes by relying on a combined information theory method and classifier predictive

performance. The approach also employs an SA optimization method to save energy by adjusting the sampling node sampling rate and transmission interval. Labeled sensor data are also used to make informed decisions regarding the network's EE options.

William et al. [40] analyze a methodology of data aggregation and clustering protocols in the paper analysis of data aggregation and clustering protocol in WSNs using ML. The authors used a robust SCADA implemented with ML to optimize QoS with minimal cost. This is known as SCADA-ML. ANNs boost overall cluster head selection and formation. The residual energy, distance from a base station, and allotted bandwidth influence the cluster formation. The approach used in this algorithm eliminates extraneous data using differential entropy and forms energy-efficient clusters and data aggregation in a controlled setting. This study presents a comparative analysis of different data aggregation and clustering protocols using ML. This study provides insight into the strengths and weaknesses of other data aggregation and clustering approaches and guides future research using efficient evaluation metrics.

A comparison of limitations of the current developments in energy-efficient MAC protocols in WSNs is presented in Table 6.

In Table 7, we have also compared the energy-efficient ML protocols, focusing on the ML methods, application environment, evaluation metrics, key results, and some identified challenges.

5.4. Analysis of EE Techniques in ML-Enhanced GWSNs. EE is a critical component in developing GWSNs because it significantly reduces power consumption during data transmission, optimizes network performance, and extends the operational lifespan of the sensor node. These networks have become increasingly adaptive and intelligent by integrating ML technologies, facilitating real-time adjustments in response to dynamic network conditions and environmental factors. This section presents a comparative analysis of recent studies on

TABLE 7: Comparison of energy-efficient machine learning approaches.

Methods	Application environment	Evaluation metric	Key results	Challenges
Machine learning-based data reduction algorithm [38]	Smart agriculture	Data reduction, accuracy, energy consumption, bandwidth usage	Real temperature readings from underground sensor network. It achieves data reduction of over 70%. Improves energy efficiency and medium utilization	Performance metrics like network latency, energy consumption, and scalability are not considered
Correlation analysis, simulated annealing [39]	Smart home environment	Accuracy, recall, and F1-measure	Uses the filter-wrapper sensor selection method to identify subsets needed to maintain predictive performance. Adjusts sampling rates and transmission intervals based on machine learning	Performance is affected by changing environmental factors such as weather, time, and location. Balancing energy saving and maintaining the data quality and network performance
Clustering—K-means, data aggregation [40]	General purpose WSN	Network lifetime, energy consumption, packet delivery ratio, throughput	K-means clustering creates optimal clusters, reducing the amount of data transmitted	The impact of sensor mobility is not considered, and the complexity and computational overhead of the machine learning algorithms used in the protocol were not thoroughly analyzed

various energy-optimization techniques for GWSNs, examining their applicability and effectiveness when enhanced using ML.

In GWSNs, several optimization strategies contribute to EE, including clustering and routing protocols, data compression, and topology optimization. Each technique offers distinct advantages in terms of reducing energy consumption. However, it also presents specific challenges regarding scalability, adaptability, and compatibility with ML-based enhancements. Tables 5 and 6 compare these approaches, elucidating the methodology, key findings, limitations, suitability for integration with ML, and performance metrics related to EE, network lifetime, and data TP.

Table 8 summarizes the principal optimization techniques derived from recent studies, emphasizing the contribution of each method to EE and network lifetime enhancement. Table 6 provides a comparative performance analysis of compatibility with ML-enhanced GWSN applications.

Table 8 illustrates the effectiveness of various optimization techniques in reducing energy consumption, notably clustering with gray wolf optimization (GWO) [41] and data compression (EEDeflate) [42]. The potential of GWO can be further expanded by incorporating ML algorithms, enabling dynamic clustering and routing adjustments, thus enhancing scalability and adaptability. Additionally, topology optimization [43] plays a significant role in energy conservation, mainly when utilized with ML-driven topology control systems that dynamically modify routing paths in response to real-time network demands.

Table 9 presents a comparative analysis of the performance metrics pertinent to ML-enhanced GWSNs, including EE, network lifetime, data TP, and ML adaptability. The research

conducted by Harjula et al. (2018), which addresses multimedia WSNs, provides valuable insights into managing high-data-demand environments by utilizing optimization techniques that demonstrate significant compatibility with ML-driven approaches. Concurrently, while achieving moderate energy conservation, methodologies such as data compression [42] can enhance ML-enhanced WSNs when implemented as a preprocessing step for data reduction.

Analysis of ML-Enhanced GWSNs is as follows:

1. **EE in Data Transmission:** Achieving EE is imperative in GWSNs, particularly for ML-enhanced applications that perform frequent data analyses and transmissions. Techniques such as GWO [41] reduce energy consumption by optimizing the cluster head selection process. ML models can dynamically adjust cluster sizes and routing paths to conserve energy and enhance EE and network performance.
2. **Data Compression and Preprocessing:** Efficient data handling is crucial for ML applications in GWSNs. EEDeflate compression [42], which conserves 7%–10% of the energy, is particularly beneficial as a preprocessing step in ML-enhanced systems. This optimization reduces data flow and enables ML models to process data more efficiently.
3. **Scalability and Adaptability for ML-enhanced Applications:** Topology optimization in [43] provides a stable foundation for ML-enhanced applications by maintaining an optimal network structure. ML integration can adaptively reconfigure the network topology based on real-time data, extend the network's longevity, and optimize data communication.

TABLE 8: Comparison of energy optimization techniques for GWSNs.

Author	Optimization technique	Methodology	Key findings	Limitations
[41]	Gray wolf optimization (GWO) with enhanced CSMA/CA	Clustering and optimal routing selection	Achieved notable energy savings by reducing energy consumption during transmission through optimal cluster head selection, improving network lifetime	May face scalability issues under high-density network conditions
[42]	Data compression (EEDeflate)	Data compression during transmission	Energy savings of 7%–10% by reducing data size crucial for green WSNs handling substantial data	Compression alone limits adaptability to dynamic, ML-driven optimization
[43]	Topology optimization (WSNERRCA)	Single-hop and multihop routing	6% energy conservation by optimizing network topology, reducing data relay load	Limited adaptability without real-time optimization for ML-enhanced applications
[44]	Multimedia sensor network optimization	Architecture for multimedia data	Managed higher energy demands through optimized architecture, enabling more efficient video/audio data handling	Primarily focused on multimedia; may require additional ML integration for dynamic adaptability
[45]	Traditional optimization techniques	Data transmission optimization	Foundational improvement in energy efficiency, supporting prolonged operations in sensor networks	Lacks ML-based dynamic adjustment for variable network demands

TABLE 9: Analysis of network performance in ML-enhanced GWSNs with EE considerations.

Author	Energy efficiency (%)	Network lifetime improvement	Data throughput	Adaptability for ML-enhanced WSNs
[41]	30–40	High, optimized clustering	High	Highly compatible; GWO can be enhanced with ML for dynamic conditions
[42]	7–10	Moderate	Moderate	Suitable as a preprocessing step in ML models for data reduction
[43]	6	High, stable topology	High	Foundational; adaptable for ML integration in topology optimization
[44]	35–40	High for multimedia WSNs	High	Compatible with ML, particularly in multimedia WSN applications
[45]	~25	Moderate	Moderate	Basic applicability; lacks dynamic adaptability without ML

- Handling the High Data Demand in Multimedia WSNs: Harjula et al.'s [44] architecture for multimedia sensor networks, such as video and audio streaming, is essential for applications that require high bandwidth. ML can dynamically adjust network parameters for different media types, improving performance and EE.

Despite these advancements, achieving ultra-low-energy operations remains a significant challenge, particularly in high-data-demand applications, such as multimedia WSNs. As suggested by Rai and Asbe [45], future research should focus on integrating energy-efficient hardware and advanced algorithms to satisfy the energy-efficiency requirements of GWSNs. This aligns with the role of ML in developing adaptable and energy-efficient solutions that respond to changing network conditions. The results of these studies highlight the

considerable potential of ML-enhanced GWSNs for delivering energy-efficient, high-performance solutions for modern applications. By incorporating ML, GWSNs demonstrate enhanced adaptability, efficiency, and capacity for satisfying intricate network demands. This comprehensive analysis lays the groundwork for understanding the strengths and limitations of various energy-efficient techniques, paving the way for future investigations into ML-integrated GWSNs aimed at achieving sustainable network operations.

6. Protocol Integration and Performance Metrics for GWSNs

The essential protocols operating at different layers of WSNs, as well as their performance metrics, are discussed in this section. The integration of ML in the layers was first discussed, and

then the performance metrics were modeled to ensure the feasible implementation of GWSNs.

6.1. ML Integration Across Network Layers. This section reviews ML protocols for GWSNs that focus on operations at the physical, network, and data link layers. In the following sections, we briefly describe the approaches provided at the layers to improve spectrum allocation, EE, and network performance.

6.1.1. Physical Layer Integration. Wireless communication systems play an essential role in our interconnected society, with applications ranging from smart cities to health care. The core of these networks is the physical layer, where signals travel via a wireless medium. Integrating ML techniques at this level has enormous potential for enhancing performance, EE, and security. This paper discusses how ML algorithms can be incorporated into the physical layer to improve communication.

The physical layer consists of sensor nodes with power supply systems (i.e., battery- or solar-powered), essential components of WSNs. To optimize the energy used by these nodes, an energy-efficient element was designed for a power amplifier that considers the power supplied, received, and consumed [46]. This function can dynamically adapt to changing network conditions using ML algorithms such as regression models or neural networks, extending the network's lifetime. Reconfigurable beam-steering antennas in the physical layer can help mitigate jamming attacks [47]. These antennas change emission patterns to block jamming signals and increase system resiliency. ML techniques, such as random forests, can be trained to classify jamming signals in real-time, allowing for a faster response and dynamic adaptation. Cui and Yu [48] proposed a scalable deep reinforcement learning that showed that a global reinforcement learning agent could determine the best method for allocating spectrum and routing based on physical inputs. Ultimately, they concluded that tackling communication and networking problems that are otherwise difficult using ML is feasible.

6.1.2. Network Layer Integration. The network layer routes packets through intermediary routers and packet forwarding. The network layer tends to be the most energy-consuming overhead in the routing process. There are various ways in which a node can transmit data to a base station. The transmission depends on the structure of the network. These include flat structure routing, hierarchical structure routing, QoS-based routing, multipath routing protocols, location-based routing, query-based routing, and proactive path establishment.

ML, particularly reinforcement learning, can be incorporated into the network layer to increase EE. ML algorithms can predict or forecast optimal routing paths to decrease data duplication during data transfer. ML algorithms may also improve network resource management by analyzing data transmission trends and traffic flow. The use of ML in GWSNs not only saves energy but also increases the network lifetime. For example, reinforcement-learning algorithms can alter real-time routing protocols, resulting in a more responsive and efficient network layer. These strategies allow the network to learn from previous actions and outcomes, making more informed packet routing

decisions that save energy without sacrificing the data integrity or delivery time. Chen et al. [49] investigated ML-enabled IoT security and emphasized the relevance of ML in improving network layer operations. Lata and Mehfuz [46] studied the application of artificial intelligence and ML to WSNs. They applied reinforcement-learning techniques to improve the EE of a network.

6.1.3. Datalink Layer Integration. The data link layer, the second level in the seven-layer OSI model of computer networking, transfers data between nodes over the physical layer. One of the primary issues in this layer is the development of MAC protocols that govern how data packets are sent from one node to another to prevent collisions and maintain efficient network use. Recent advances have demonstrated the use of ML to automate and optimize the development of MAC protocols. For example, Pasandi and Nadeem [50] automated the design of MAC protocols, reduced human error, and improved network performance in various scenarios. Their AlphaMAC framework uses a Q-learning approach, reinforcement learning, to develop efficient MAC protocols automatically. AlphaMAC has proven its capacity to choose the most effective components from the ALOHA protocol-building block set, resulting in a protocol outperforming the regular ALOHA protocol. Uddin et al. [36] presented PSE²-MAC, an energy-efficient MAC protocol for asynchronous WSNs based on predictive scheduling. This protocol handles the frequent wireless communication challenges of idle listening and beacon collisions, which waste energy. PSE²-MAC proved to work better due to the integration of ML to improve network efficiency and stability.

6.2. Performance Evaluation Metrics. Performance metrics are a crucial part of ML pipelines. They help researchers determine whether they are making progress and quantifying it. All ML models, whether linear regression or state-of-the-art techniques such as bidirectional encoder representations from transformers (BERTs), require a metric to judge performance. Every ML task can be broken down into regression or classification [51]. There are several metrics for both problems. The popular ones discussed in this paper include the mean absolute error (MAE), mean squared error (MSE), root MSE (RMSE), R-squared (R^2) for regression problems, and accuracy, precision, recall, F1 score, and area under the receiver operating characteristic (ROC) curve (AUC) for classification problems [52].

6.2.1. Regression Metrics. They are used to evaluate the performance of regression models with a continuous output. These metrics are based on calculating the distance between the predicted and ground truth values. Some standard regression metrics include the following.

- **MSE:** The average squared difference between the predicted and actual values is calculated.
- **MAE:** Calculates the average absolute difference between the predicted and actual values.
- **RMSE:** The square root of the average squared difference between the predicted and actual values was calculated using this metric.

- R^2 : This metric assesses how well the regression model fits the data by computing the proportion of variance in the dependent variable that can be predicted by the independent variable(s).

6.2.2. Classification Metrics. They were used to evaluate the performance of classification models with discrete outputs. These metrics are based on calculating the similarity or agreement between predicted and ground-truth class labels. Some standard classification metrics include the following:

- **Accuracy:** This metric calculates the proportion of classified instances from the total number.
- **Precision:** This metric calculates the proportion of true-positive instances from the total number of positive instances predicted by the model.
- **Recall:** This metric computes the fraction of real positive examples among the total number of positive instances.
- **F1 Score:** This metric calculates the harmonic mean of the precision and recall.
- **AUC:** The ROC compares the true positive rate to the false positive rate at various categorization criteria.

6.2.3. Network Lifetime. In WSNs, network lifetime is defined as the time between the start of network operations and the first node running out of energy. This is an essential metric for determining the long-term viability and dependability of GWSNs. ML algorithms can assist in extending the network lifespan by improving energy consumption and sensor node harvesting [53].

6.2.4. Scalability. Scalability is the network's ability to accommodate many nodes while maintaining performance and QoS. This is an essential statistic for assessing the adaptability and flexibility of GWSNs. ML approaches can help increase the scalability of GWSNs by modifying the network characteristics and protocols based on network size and dynamics [54].

6.2.5. Security. In WSNs, security is the degree to which a network is protected against malicious assaults and unauthorized access. This is a critical statistic for ensuring network data and services' confidentiality, integrity, and availability. ML approaches can improve GWSN security by identifying and blocking various attacks, including jamming, eavesdropping, spoofing, and denial of service [55, 56].

6.2.6. Overhead. The overhead is the amount of additional data or control information sent along with actual data. It is an essential indicator for assessing network communication, routing efficiency, and overhead. By compressing and aggregating data and minimizing control messages, ML approaches can assist in reducing the overhead of GWSNs [57].

6.2.7. Robustness. Robustness is the ability of ML-based GWSNs to cope with network uncertainties, noises, and errors. The resilience and fault tolerance of ML models and network protocols can be used to determine robustness. Robustness is

essential to ensure the stability and security of ML-based GWSNs.

6.3. Performance Metrics in ML-Based GWSNs. This section analyzes critical performance metrics for GWSNs and how their efficiency can be enhanced by integrating ML techniques. Specifically, we focus on three critical metrics: EE, TP, and delay reduction (DR).

6.3.1. EE. The EE of GWSNs measures how effectively the network conserves energy while transmitting data. The refined equation for EE incorporates ML's role of ML in prioritizing and classifying data and reducing energy consumption by focusing on the most critical transmissions in Equation (1).

$$EE = \frac{\text{Data}_{\text{trans}} \times W_{\text{data}}}{\text{Energy}_{\text{tx}} + \text{Energy}_{\text{rx}} + \text{Energy}_{\text{idl}} + \text{Energy}_{\text{ML}}}. \quad (1)$$

- $\text{Data}_{\text{trans}}$: Total amount of useful data transmitted across the network.
- W_{data} : Priority weights assigned by ML algorithms that classify data based on their importance.
- $\text{Energy}_{\text{tx}}, \text{Energy}_{\text{rx}}, \text{Energy}_{\text{idl}}$: Energy consumed during data transmission, reception, and idling periods, respectively.
- $\text{Energy}_{\text{ML}}$: Additional energy ML processes consume, such as data processing and decision-making.

ML's capacity to optimize these factors leads to substantial energy savings, with studies indicating a 50% reduction in energy consumption compared to traditional WSNs.

6.3.2. TP. TP is a crucial metric for successfully delivering data packets within a network over time. In ML-based GWSNs, the TP equation is refined to account for the prioritization of data packets and the additional processing time required by ML algorithms, as indicated in Equation (2).

$$TP = \left(\sum_{i=1}^N (i=1)^N + \frac{\text{Pkt}_{\text{dl},i} + \text{Pr}_{\text{data},i}}{Tt + T_{\text{ML}}} \right). \quad (2)$$

- $\text{Pkt}_{\text{dl},i}$: Number of successfully delivered packets for Type III data.
- $\text{Pr}_{\text{data},i}$: Priority level assigned to each data TYPE III by the ML algorithms.
- Tt : Total time for data transmission.
- T_{ML} : Additional time incurred owing to ML processing.

By dynamically managing data priorities and routing, ML-enhanced GWSNs can achieve up to 30% improvement in TP.

6.3.3. DR. A delay or latency in data transmission is critical, particularly in applications that require timely data delivery. The delay DDD in a GWSN can be expressed as the sum of several delay components, including ML-related delays, as shown in Equation (3).

$$D = \sum_{i=1}^N \left(\frac{\text{Data}_{\text{size},i}}{B_{\text{alloc},i}} + \text{PD}_i + \text{PrD}_i + \text{QD}_i + \text{MLTd}_i \right). \quad (3)$$

- $\text{Data}_{\text{size},i}$: The size of the data packet i .
- $\text{Bandwidth}_{\text{allocated},i}, B_{\text{alloc},i}$: Bandwidth allocated to data type i , which ML algorithms can optimize.
- $\text{Propagationdelay}_i, \text{PD}$: Time taken for data to travel from source to destination.
- $\text{Processingdelay}_i, \text{PrD}$: Time used to process data packets at nodes.
- $\text{QueuingDelay}_i, \text{QD}$: The time packets spend waiting in queues before being processed or transmitted.
- $\text{MLDecisionTime}_i, \text{MLTd}_i$: Additional time required for ML algorithms to make decisions regarding data routing and prioritization.

The DR achieved by ML-based enhancements is calculated by comparing the delay in conventional networks, D_{conv} to the delay in ML-based networks, $D_{\text{ML-based}}$:

$$\text{DR} = \left(\frac{D_{\text{conv}} - D_{\text{ML-based}}}{D_{\text{conv}}} \right) \times 100\%. \quad (4)$$

Equation (4) shows the percentage reduction in delay owing to the application of the ML algorithms. Empirical studies have indicated that ML-enhanced GWSNs can achieve a 40% reduction in delay due to more efficient routing, data prioritization, and dynamic bandwidth allocation. The equations presented in this section underscore the transformative role of ML in optimizing GWSNs. ML techniques enhance EE, TP, and DR by making GWSNs more effective and sustainable. The reported improvements, such as a 50% reduction in energy consumption, a 30% increase in TP, and a 40% reduction in delay, are supported by empirical data, demonstrating the significant potential of ML in advancing the capabilities of GWSNs.

7. Implementations of ML in GWSNs

This paper provides examples of ML implementations for GWSNs in various domains, such as smart cities, healthcare, agriculture, and military. It also identifies the future research directions and opportunities for ML-based GWSNs. The application areas include agriculture, water quality, air pollution, animal tracking, and underground mining. The following sections describe these application areas and the characteristics of ML in the application domain.

7.1. ML Agriculture Applications. Over the past decade, the agricultural sector has received many novel inputs to automate processes involved in agricultural production. Some of the focus areas have been forecasting soil greenhouse gas emissions from an agricultural field [58], reviewing ML models in agriculture from an economist's perspective [59], and exploiting datasets for decision-making [60]. Of the selected articles on this topic, few concentrate on GWSNs but focus solely on

applying ML techniques in agriculture. Table 10 summarizes the selected papers on agricultural applications using ML.

7.2. ML Water Quality Monitoring Applications. Poor water quality can pose health risks to both humans and ecosystems. Several factors, such as turbidity, bacteria level, salt, and dissolved oxygen concentration, can be used to measure water quality. Automating the monitoring of these parameters can significantly affect the environment.

The importance of water quality has drawn the attention of many researchers in this domain. We concentrated on automating the processes and activities involved in monitoring and calculating parameters contributing to water quality. A survey by Dogo et al. [65] proved that there has been sparse exploitation of applications in this domain and suggested a new framework called hybrid DL-ELM for detecting water anomalies.

A recent study by Ilić et al. [66] using the Naïve Bayes algorithm predicted 64 out of 68 cases of water anomalies. Given these, researchers should pay attention to applications for monitoring water quality using ML techniques. Table 11 summarizes the recent ML applications for water quality monitoring.

7.3. ML for Animal Tracking Applications. As agriculture feeds the entire population, it is vital to the nation's economy. The need to hire additional workers to help with planting and farm animal care is common on large farms. Hence, automating animal monitoring and tracking is essential if livestock and poultry constitute a subset of agriculture.

In work done by Nobrega et al. [69], they proposed an animal behavior monitoring platform with a local network to gather animal data and a cloud platform with processing and storage capabilities to shepherd ovines autonomously within vineyard areas. They discovered that the decision tree and random forest models were slightly more accurate than the other compared models. Bonneau et al. [70] concluded that image analysis could be an interesting alternative to GPS with comparable accuracy and significantly lower cost. Table 12 summarizes recent ML applications for animal tracking.

7.4. ML for Air Pollution Applications. Air pollution is a critical problem in urban and industrial areas. Several researchers have addressed this problem, and some contributions, including ML, are presented in Table 13.

7.5. ML in Underground Mining Applications. Mining is another ML application. Ren et al. [74] did a study in which they proposed a framework that predicts surface cracks and deformation due to underground mining. Table 14 summarizes some recent ML applications for underground monitoring.

8. Open Issues and Future Research Directions

8.1. Open Issues. GWSNs are promising technologies for various applications that require EE and environmental sustainability. However, GWSNs face many challenges that limit their performance and potential. Several challenges and open research issues in GWSNs require further research.

TABLE 10: Characteristics of ML in agriculture applications.

References	ML approach used	Category	Tools	Key results	Simulation/real tested
[61]	Unsupervised learning (decision tree, k-nearest neighbor, support vector machine)	Design	N/A	(92.5% accuracy) Wheat planting dates could be determined successfully from climate information obtained in the first 300 days with the help of machine learning techniques combined with feature selection using a genetic algorithm, which will prevent low productivity and financial and labor loss due to inaccurate planting dates	Simulation
[62]	Supervised learning	Design	Apache Spark data frame, Scikit-learn, Google Collaboratory, Google <i>Dataprocluster</i>	Feature selection counts for potatoes show that soil water holding capacity was always selected. Early season predictions using yield trend indicated that the baseline could make early season predictions better than the null method	Real tested
[59]	Unsupervised learning with neural networks	Design	RapidMiner Studio	Results demonstrate that there is plenty of room for innovation while supporting requests and needs from businesses looking to employ an agricultural, industrial business that is sustainable and optimized, investing in technology and the knowledge and skilled workforce needed to get the most out of it	Simulation
[63]	Unsupervised learning (artificial neural networks and random forest)	Design	1. Sentinel-2 2. Landsat 8	The model successfully estimates GPP across various C3 crop types and environmental conditions, using no local information from the corresponding sites	Simulation
[64]	• Supervised learning • Unsupervised learning • Reinforcement learning • Federated learning	Review	Systematic review	Supervised and unsupervised learning have been mainly used for precision irrigation with positive outcomes	None

TABLE 11: Characteristics of ML water quality monitoring applications.

References	ML approach used	Category	Tools	Key results	Simulation/real tested
[67]	• Artificial neural networks • Support vector machine • Linear regression	Design	• RapidMiner Studio • Azure Machine Learning Studio • IBM SPSS Modeler • WEKA	The most accurate model among the single ANNs model, the ANNs tiering model, and the MetaFA-LSSVR hybrid model were identified and used to predict the CTSI of water quality in reservoirs	Simulation
[66]	Naïve bayes algorithm	Design	• Netica Software • Water quality index	Predicted the class of water sample in 64 out of 68 cases	Simulation
[65]	• Extreme learning machine • Neural networks	Review	N/A	• In terms of accuracy in feature learning and a reduction in false positive rates, deep learning approaches perform better than conventional machine learning techniques. However, because various datasets, models, and parameters were used throughout the studies, it is challenging to compare them fairly. • Although the extreme learning machine (ELM) has made strides and has many benefits, its use in this field is still underutilized	None
[68]	• Extreme learning machine • Recurrent neural networks • Generalized regression neural networks • An adaptive neuro-fuzzy inference system • Support vector machine • Decision tree	Review	Comparative analysis	Deep learning techniques are suitable for univariate and multivariate time series forecasting problems, such as predicting water quality	None

TABLE 12: Characteristics of ML for animal tracking applications.

References	ML approach used	Category	Tools	Key results	Simulation/real testbed
[71]	• Regression algorithm • K-nearest neighbor • Random forest • Support vector machine • Neural network	Review	Systematic review	It was discovered that the application of ML to PLF is still in its infancy and presents several research hurdles	None
[69]	• Random forest • Decision tree • XGBoost • K-nearest neighbor • SVM • Naïve Bayes	Review	SheepIT	Random forest and decision tree had the highest accuracy of 9.693, then the other ML approaches presented	None
[70]	Deep learning	Design	Time-lapse camera	Using computer vision for animal tracking is also possible outdoors, not indoors, with top-view cameras	Real testbed

TABLE 13: Characteristics of ML for air pollution applications.

References	ML approach used	Category	Tools	Key results	Simulation/real testbed
[72]	Advanced optimization algorithm (Frobenius norm regularization)	Design	N/A	The parameter-reducing formulations and consecutive-hour-related regularisations perform better than standard regression models and regularisations	Simulation
[73]	• Support vector machine • Artificial neural networks	Design	Online air quality prediction system	Results demonstrate a remarkable speedup using the online technique while maintaining the SVM classifier's accuracy. The hourly predictions for the following 24 h indicate an overall accuracy of 0.71, a root mean square error of 0.54, and a coefficient of determination of 0.81 compared to the measured pollution data from Tehran pollution monitoring stations	Simulation

TABLE 14: Characteristics of ML in underground mining.

References	ML approach used	Category	Tools	Key results	Simulation/real testbed
[74]	• K-means • Dynamic time warping • Association rule mining • Gray relation analysis • Support vector regression	Design	N/A	Results demonstrate that the anticipated value and the measured data agree well	Real testbed
[75]	Support vector machine (SVM) Artificial neural networks	Design	N/A	The results also indicated that H , L_b , t , and α are the most influential parameters that should be used to predict rock mass displacement in tunnels and underground mines	Simulation

8.1.1. Energy Management. GWSNs depend on limited and renewable energy sources, including solar, wind, and vibration, to power sensor nodes. However, these energy sources can be unpredictable, intermittent, and heavily influenced by environmental conditions. Future research should focus on improving the reliability and predictability of these energy

sources to enhance the performance and efficiency of GWSNs. Therefore, GWSNs must manage the energy harvesting, storage, and consumption of nodes to ensure continuous and reliable operation. Moreover, GWSNs must balance the trade-off between EE and QoS, such as TP, delay, reliability, and security.

8.1.2. Network Topology and Routing. GWSNs have dynamic and heterogeneous network topologies, where nodes may have different capabilities, roles, and mobility patterns. Moreover, GWSNs may have large-scale and dense deployments, where the nodes must communicate with each other and with multiple sinks or gateways. Therefore, GWSNs must design scalable and adaptive network structures and routing protocols that address network dynamics, heterogeneity, and complexity. Furthermore, GWSNs must optimize routing paths, data aggregation, and fusion processes to reduce communication overhead and energy consumption.

8.1.3. Data Processing and Analysis. GWSNs generate massive amounts of data from various sensors and applications that must be processed and analyzed promptly and efficiently. However, GWSNs have limited computational and storage resources, which makes performing complex and high-dimensional data processing and analysis tasks challenging. Therefore, GWSNs must leverage distributed, collaborative, and online learning techniques to handle data volume, velocity, and variety. Moreover, GWSNs must extract valuable and actionable information and knowledge from data to support decision-making and feedback control.

8.2. Future Research Directions. Although ML techniques have been extensively applied in WSNs, several unresolved challenges still require further research. ML is a potent instrument that can offer solutions to these persisting issues, particularly in GWSNs. ML has the potential to empower GWSNs to learn from data, environmental conditions, and network behaviors, thereby enabling them to adapt to and optimize their performance. Future research will explore various ways ML can tackle these issues in GWSNs.

8.2.1. Management of Resources. As energy consumption is a critical concern in WSNs, future research will focus on enhancing communication-related protocols and developing a framework to optimize bandwidth and power for WSNs. These protocols will incorporate efficient MAC protocol design techniques and ML methods to address node redundancy due to ML optimization. ML will enable GWSNs to predict energy availability and demand and adjust energy harvesting, storage, and consumption strategies. It will also forecast solar irradiance and wind speed and optimize the nodes' duty cycling and transmission power. Furthermore, ML can assist GWSNs in balancing EE and QoS trade-offs by learning the optimal parameters and policies to achieve the desired QoS levels with minimal energy consumption. Researchers will continue to explore these areas to improve the efficiency and effectiveness of WSNs.

8.2.2. Distributed ML Techniques. As WSNs increase, the amount of data they produce over a certain period also increases. As such, traditional ML techniques face scalability issues and accurately predict data. Distributed ML uses multi-node algorithms and systems that scale up to enormous input data sizes, boost accuracy, and improve performance. These algorithms are more efficient than conventional algorithms because they can be fed massive amounts of data, dramatically

reducing learning errors. Each node in a sensor network can be equipped with an adaptive ML model.

8.2.3. Cross-Layer Protocols. Future research will focus on improving the determination of the energy value in routing packets, which requires communication between the layers. The energy value in packet routing assists the routing protocol in selecting energy-efficient routes. The development and enhancement of cross-layer design will be a crucial area of study because it facilitates the creation of route energy packets that transfer energy values between nodes. Researchers will also expand their focus to include the future of GWSNs and explore how these protocols can be integrated and optimized within these networks. *The cross-layer protocols* in GWSNs are worth exploring for further development.

8.2.4. Feature Extraction, Data Processing, and Analysis. Future research will address the current bottleneck of ML models, which is their limitation in learning only the features of the data they have been trained with. Deep learning (DL) that focuses on feature learning is essential. ML can aid GWSNs in performing efficient and effective data processing and analysis tasks by leveraging distributed, collaborative, and online learning techniques. Researchers will explore how ML can compress, filter, and classify data at the node level and aggregate, fuse, and mine data at the cluster or network level. Additionally, they will investigate how ML can help GWSNs extract valuable and actionable information and knowledge from data by learning data patterns, trends, and anomalies that can support decision-making and feedback control. Feature extraction, data processing, and analysis require further exploration.

8.2.5. Routing Using ML Techniques. Future research will focus on reducing the energy consumption in packet transmission from source nodes to destination nodes. ML models, including ANNs, Bayesian-based routing, and reinforcement learning, will be investigated in detail in future studies. The goal is to identify the most efficient routes to extend the network's lifetime. The potential of ML to enable GWSNs to understand network topology and node characteristics and adapt the network structure and routing protocol accordingly is a crucial study area. Researchers will also investigate how ML can cluster nodes based on their location, energy, and role and select cluster heads and gateways to facilitate communication and data aggregation. Optimizing routing paths and data fusion processes through ML will also be examined, focusing on learning optimal routes and data correlations to minimize communication costs and maximize data quality.

9. Conclusion

WSNs exhibit distinctive characteristics compared with traditional networks, necessitating specialized protocols and tools to address their unique challenges and constraints. These networks require novel approaches in various domains, including routing, security, scheduling, localization, node clustering, data aggregation, fault detection, and data integrity. Notably, these solutions must be energy-conscious and real-time. ML emerges as a powerful paradigm to enhance the capabilities of WSNs, enabling them to adapt dynamically to their immediate

environment. Researchers have increasingly leveraged ML techniques to address WSN challenges. DL techniques have become prominent and are characterized using deep neural networks. DL methods can extract high-level features from unprocessed sensor data, providing advantages such as decreased computational complexity, increased feasibility for identifying optimal solutions and improved EE. Our comprehensive review of related studies from 2018 to 2022 revealed a consistent trend: researchers favor neural networks as the most effective ML technique, yielding highly accurate results. However, an essential consideration lies in the quality and quantity of the training data. The success of ML models hinges on the judicious selection and preparation of data, which significantly influences the model's performance. In summary, integrating ML techniques holds immense promise for advancing WSNs. However, careful attention to data quality and model training remains critical for optimal outcomes. Future research should focus on developing robust, adaptive, and scalable ML solutions tailored to resource-constrained and dynamic network environments. It will be beneficial to focus on developing lightweight scalable and distributed ML solutions, ensuring data privacy and reducing computational overhead. Reinforcement learning for real-time network optimization in GWSN and secure ML techniques such as differential privacy and homomorphic encryption are interesting research ideas for future works.

Data Availability Statement

No data are available for this manuscript.

Conflicts of Interest

The authors declare no conflicts of interest.

Funding

The authors declare that no funds, grants, or other support were received during the preparation of this manuscript.

References

- [1] K. S. Adu-Manu, J.-D. Abdulai, F. Engmann, et al., "WSN Architectures for Environmental Monitoring Applications," *International Journal of Distributed Sensor Networks* 2022 (2022): 7823481, 18.
- [2] S. Bilgen, "Structure and Environmental Impact of Global Energy Consumption," *Renewable and Sustainable Energy Reviews* 38 (2014): 890–902.
- [3] D. L. Cahya, "Analysis of Urban Agriculture Sustainability in Metropolitan Jakarta (Case Study: Urban Agriculture in Duri Kosambi)," *Procedia - Social and Behavioral Sciences* 227 (2016): 95–100.
- [4] F. Engmann, F. A. Katsriku, J.-D. Abdulai, K. S. Adu-Manu, and F. K. Banaseka, "Prolonging the Lifetime of Wireless Sensor Networks: A Review of Current Techniques," *Wireless Communications and Mobile Computing* 2018 (2018): 8035065, 23.
- [5] K. S. Adu-Manu, N. Adam, C. Tapparelo, H. Ayatollahi, and W. Heinzelman, "Energy-Harvesting Wireless Sensor Networks (EH-WSNs) A Review," *ACM Transactions on Sensor Networks (TOSN)* 14, no. 2 (2018): 1–50.
- [6] R. Ahmad, W. Alhasan, R. Wazirali, and N. Aleisa, "Optimization Algorithms for Wireless Sensor Networks Node Localization: An Overview," *IEEE Access* 12 (2024): 50459–50488.
- [7] R. Priyadarshi, "Exploring Machine Learning Solutions for Overcoming Challenges in IoT-Based Wireless Sensor Network Routing: A Comprehensive Review," *Wireless Networks* 30, no. 4 (2024): 2647–2673.
- [8] F. Engmann, K. S. Adu-Manu, J.-D. Abdulai, and F. A. Katsriku, "Network Performance Metrics for Energy Efficient Scheduling in Wireless Sensor Networks (WSNs)," *Wireless Communications and Mobile Computing* 2021 (2021): 9635958, 14.
- [9] P. Singh, "Enhancing Performance of Hybrid Electric Vehicle Using Optimized Energy Management Methodology," *International Journal of Data Informatics and Intelligent Computing* 2, no. 3 (2023): 1–10.
- [10] M. Deb and S. Roy, "Enhanced-Pro: A New Enhanced Solar Energy Harvested Prediction Model for Wireless Sensor Networks," *Wireless Personal Communications* 117, no. 2 (2021): 1103–1121.
- [11] D. K. Sah, A. Hazra, R. Kumar, and T. Amgoth, "Harvested Energy Prediction Technique for Solar-Powered Wireless Sensor Networks," *IEEE Sensors Journal* 23, no. 8 (2023): 8932–8940.
- [12] S. Sarang, G. M. Stojanović, M. Drieberg, S. Stankovski, K. Bingi, and V. Jeoti, "Machine Learning Prediction Based Adaptive Duty Cycle MAC Protocol for Solar Energy Harvesting Wireless Sensor Networks," *IEEE Access* 11 (2023): 17536–17554.
- [13] I. Ahmed, M. Ahmad, J. J. P. C. Rodrigues, G. Jeon, and S. Din, "A Deep Learning-Based Social Distance Monitoring Framework for COVID-19," *Sustainable Cities and Society* 65 (2021): 102571.
- [14] M. A. Alsheikh, S. Lin, D. Niyato, and H.-P. Tan, "Machine Learning in Wireless Sensor Networks: Algorithms, Strategies, and Applications," *IEEE Communications Surveys & Tutorials* 16, no. 4 (2014): 1996–2018.
- [15] B. Mahesh, "Machine Learning Algorithms—A Review," *International Journal of Science and Research (IJSR)* 9, no. 1 (2020): 381–386.
- [16] N. Yamin and G. Bhat, "Online Solar Energy Prediction for Energy-Harvesting Internet of Things Devices," in *2021 IEEE/ACM International Symposium on Low Power Electronics and Design (ISLPED)*, (Boston, MA, USA: IEEE, 2021), 1–6.
- [17] S. Shresthamali, M. Kondo, and H. Nakamura, "Adaptive Power Management in Solar Energy Harvesting Sensor Node Using Reinforcement Learning," *ACM Transactions on Embedded Computing Systems* 16, no. 5s (2017): 1–21.
- [18] M. S. Ali, M. K. Islam, A. A. Das, D. U. S. Duranta, M. F. Haque, and M. H. Rahman, "A Novel Approach for Best Parameters Selection and Feature Engineering to Analyze and Detect Diabetes: Machine Learning Insights," *BioMed Research International* 2023 (2023): 8583210, 15.
- [19] A. Sharma, A. Jain, P. Gupta, and V. Chowdary, "Machine Learning Applications for Precision Agriculture: A Comprehensive Review," *IEEE Access* 9 (2021): 4843–4873.
- [20] F. Engmann, F. A. Katsriku, J.-D. Abdulai, and K. S. Adu-Manu, "Reducing the Energy Budget in WSN Using Time Series Models," *Wireless Communications and Mobile Computing* 2020 (2020): 8893064, 15.
- [21] A. Sarmadi, M. Hassanzadeganroudsari, and M. Soltani, "Artificial Intelligence and Machine Learning Applications in Vaccine Development," in *Bioinformatics Tools for Pharmaceutical Drug Product Development*, eds. V. Chavda, K. Anand, and V. Apostolopoulos, (2023): 233–253.

- [22] M. Mittal, R. P. de Prado, Y. Kawai, S. Nakajima, and J. E. Muñoz-Expósito, "Machine Learning Techniques for Energy Efficiency and Anomaly Detection in Hybrid Wireless Sensor Networks," *Energies* 14, no. 11 (2021): 3125.
- [23] M. Pundir and J. K. Sandhu, "A Systematic Review of Quality of Service in Wireless Sensor Networks Using Machine Learning: Recent Trend and Future Vision," *Journal of Network and Computer Applications* 188 (2021): 103084.
- [24] S. V. N. Sreenivasu, T. Sathesh Kumar, O. Bin Hussain, A. R. Yeruva, S. R. Kabat, and A. Chaturvedi, "Cloud-Based Electric Vehicle's Temperature Monitoring System Using IOT," *Cybernetics and Systems* (2023): 1–16.
- [25] K. S. Adu-Manu, *A Study into Lifetime Maximization of Wireless Sensor Networks for Water Quality Monitoring* (University of Ghana, 2019).
- [26] K. S. Adu-Manu, F. Katsriku, J.-D. Abdulai, J. M. Gómez, and W. Heinzelmann, "Network Lifetime Maximization with Adjustable Node Transmission Range," in *Smart Cities/Smart Regions—Technische, Wirtschaftliche Und Gesellschaftliche Innovationen*, eds. J. Marx Gómez, A. Solsbach, T. Klenke, and V. Wohlgemuth, (Springer Vieweg, Wiesbaden, 2019): 693–707.
- [27] M. Kulin, T. Kazaz, E. De Poorter, and I. Moerman, "A Survey on Machine Learning-Based Performance Improvement of Wireless Networks: PHY, MAC and Network Layer," *Electronics* 10, no. 3 (2021): 318.
- [28] H. Rhim, K. Tamine, R. Abassi, D. Sauveron, and S. Guemara, "A Multi-Hop Graph-Based Approach for an Energy-Efficient Routing Protocol in Wireless Sensor Networks," *Human-Centric Computing and Information Sciences* 8 (2018): 30.
- [29] C. Xu, Z. Xiong, G. Zhao, and S. Yu, "An Energy-Efficient Region Source Routing Protocol for Lifetime Maximization in WSN," *IEEE Access* 7 (2019): 135277–135289.
- [30] V. Pandiyaraju, R. Logambigai, S. Ganapathy, and A. Kannan, "An Energy Efficient Routing Algorithm for WSNs Using Intelligent Fuzzy Rules in Precision Agriculture," *Wireless Personal Communications* 112, no. 1 (2020): 243–259.
- [31] M. Ahmed, M. Salleh, and M. I. Channa, "CBE2R: Clustered-Based Energy Efficient Routing Protocol for Underwater Wireless Sensor Network," *International Journal of Electronics* 105, no. 11 (2018): 1916–1930.
- [32] R. A. Khan, S. Khan, A. Ullah, F. Arif, and T. A. Alghamdi, "An Energy-Efficient Routing Protocol for Wireless Body Area Sensor Networks," *Wireless Personal Communications* 99, no. 4 (2018): 1443–1454.
- [33] G. Mehmood, M. Z. Khan, A. K. Bashir, Y. D. Al-Otaibi, and S. Khan, "An Efficient QoS-Based Multi-Path Routing Scheme for Smart Healthcare Monitoring in Wireless Body Area Networks," *Computers and Electrical Engineering* 109, no. PA (2023): 108517.
- [34] A. Chehri, "Energy-Efficient Modified DCC-MAC Protocol for IoT in E-Health Applications," *Internet of Things* 14 (2021): 100119.
- [35] Y. Liu, Z. Zheng, X. Liu, Z. Jia, and L. Hanzo, "QTSAC: An Energy-Efficient MAC Protocol for Delay Minimization in Wireless Sensor Networks," *IEEE Access* 6, no. Part II (2018): 8273–8291.
- [36] M. N. Uddin, M. O. Rahman, and S. Kazary, "A Predictive Schedule-Based Energy Efficient MAC Protocol for Wireless Sensor Networks," in *2022 5th International Conference on Computing and Informatics (ICCI)*, (New Cairo, Cairo, Egypt: IEEE, 2022), 424–429.
- [37] P. Lin, C. Qiao, and X. Wang, "Medium Access Control With a Dynamic Duty Cycle for Sensor Networks," in *2004 IEEE Wireless Communications and Networking Conference (IEEE Cat. No. 04TH8733)*, 3, (Atlanta, GA, USA: IEEE, 2004), 1534–1539.
- [38] C. Salim and N. Mitton, "Machine Learning-Based Data Reduction in WSN for Smart Agriculture," *Advances in Intelligent Systems and Computing* 1151 (2020): 127–138.
- [39] J. Kang, J. Kim, M. Kim, and M. Sohn, "Machine Learning-Based Energy-Saving Framework for Environmental States-Adaptive Wireless Sensor Network," *IEEE Access* 8 (2020): 69359–69367.
- [40] P. William, A. Badholia, V. Verma, A. Sharma, and A. Verma, "Analysis of Data Aggregation and Clustering Protocol in Wireless Sensor Networks Using Machine Learning," in *Evolutionary Computing and Mobile Sustainable Networks*, eds. V. Suma, X. Fernando, K. L. Du, and H. Wang, (Springer, Singapore, 2022): 925–939.
- [41] M. Kaddi, M. Omari, K. Salameh, and A. Alnoman, "Energy-Efficient Clustering in Wireless Sensor Networks Using Grey Wolf Optimization and Enhanced CSMA/CA," *Sensors* 24, no. 16 (2024): 5234.
- [42] P. D. Ganjewar, S. Barani, and S. J. Wagh, "Energy Efficient Deflate (EEDeflate) Compression for Energy Conservation in Wireless Sensor Network," in *Intelligent Systems Technologies and Applications 2016*, eds. J. Corchado Rodriguez, S. Mitra, S. Thampi, and E. S. El-Alfy, (Springer, Cham, 2016): 287–296.
- [43] S. S. Akende, M. A. Ahaneke, U. N. Nwawelu, U. C. Amazue, and D. A. Amoke, "Improving Energy Efficiency of Wireless Sensor Networks through Topology Optimization," *International Journal of Emerging Technology and Advanced Engineering* 12, no. 12 (2022): 107–116.
- [44] E. Harjula, T. Mekonnen, M. Komu, et al., "Energy Efficiency in Wireless Multimedia Sensor Networking: Architecture, Management and Security," in *Greening Video Distribution Networks*, ed. A. Popescu, (Springer, Cham, 2018): 133–157.
- [45] A. K. Rai and C. Asbe, "Reduction of Energy Consumption During Data Communication Evaluated via Optimization Technique for WSN," in *2023 IEEE 4th Annual Flagship India Council International Subsections Conference (INDISCON)*, (Mysore, India: IEEE, 2023), 1–8.
- [46] S. Lata and S. Mehruz, "Machine Learning Based Energy Efficient Wireless Sensor Network," in *2019 International Conference on Power Electronics, Control and Automation (ICPECA)*, (New Delhi, India: IEEE, 2019), 1–5.
- [47] M. Jacovic, X. R. Rey, G. Mainland, and K. R. Dandekar, "Mitigating RF Jamming Attacks at the Physical Layer With Machine Learning," *IET Communications* 17, no. 1 (2023): 12–28.
- [48] W. Cui and W. Yu, "Scalable Deep Reinforcement Learning for Routing and Spectrum Access in Physical Layer," *IEEE Transactions on Communications* 69, no. 12 (2021): 8200–8213.
- [49] Z. Chen, Y. Zhang, Y. Li, Y. Wang, X. Liu, and A. V. Vasilakos, "Machine Learning-Enabled IoT Security: Open Issues and Challenges Under Advanced Persistent Threats," *ACM Computing Surveys* 55, no. 5 (2022): 1–37.
- [50] H. Barahouei Pasandi and T. Nadeem, "Poster: Towards Self-Managing and Self-Adaptive Framework for Automating MAC Protocol Design in Wireless Networks," in *HotMobile '19: Proceedings of the 20th International Workshop on Mobile Computing Systems and Applications*, (New York, NY, United States: Association for Computing Machinery, 2019).

- [51] M. S. Alsahli, M. M. Almasri, M. Al-Akhras, A. I. Al-Issa, and M. Alawairdhi, "Evaluation of Machine Learning Algorithms for Intrusion Detection System in WSN," *International Journal of Advanced Computer Science and Applications* 12, no. 5 (2021): 617–626.
- [52] J. Birla, "Performance Metrics in Wireless Sensor Network," *International Journal of Engineering Research & Technology (IJERT)* 3, no. 10 (2015): 1–4.
- [53] Q. Zhang, D. Feng, F. Wang, and Y. Xie, "An Efficient, QoS-Aware I/O Scheduler for Solid State Drive," in *2013 IEEE 10th International Conference on High Performance Computing and Communications & 2013 IEEE International Conference on Embedded and Ubiquitous Computing*, (Zhangjiajie, China: IEEE, 2013), 1408–1415.
- [54] S. Misra, S. Chatterjee, and M. S. Obaidat, "On Theoretical Modeling of Sensor Cloud: A Paradigm Shift From Wireless Sensor Network," *IEEE Systems Journal* 11, no. 2 (2017): 1084–1093.
- [55] A. L. Buczak and E. Guven, "A Survey of Data Mining and Machine Learning Methods for Cyber Security Intrusion Detection," *IEEE Communications Surveys & Tutorials* 18, no. 2 (2016): 1153–1176.
- [56] K. S. Adu-Manu, F. Engmann, G. Sarfo-Kantanka, G. E. Baiden, and B. A. Dulemordzi, "WSN Protocols and Security Challenges for Environmental Monitoring Applications: A Survey," *Journal of Sensors* 2022 (2022): 1628537, 21.
- [57] R. Amiri, H. Mehrpouyan, L. Fridman, R. K. Mallik, A. Nallanathan, and D. Matolak, "A Machine Learning Approach for Power Allocation in HetNets Considering QoS," in *2018 IEEE International Conference on Communications (ICC)*, (Kansas City, MO, USA: IEEE, 2018), 1–7.
- [58] A. Hamrani, A. Akbarzadeh, and C. A. Madramootoo, "Machine Learning for Predicting Greenhouse Gas Emissions From Agricultural Soils," *Science of the Total Environment* 741 (2020): 140338.
- [59] F. Balducci, D. Impedovo, and G. Pirlo, "Machine Learning Applications on Agricultural Datasets for Smart Farm Enhancement," *Machines* 6, no. 3 (2018): 38.
- [60] H. Storm, K. Baylis, and T. Heckeley, "Machine Learning in Agricultural and Applied Economics," *European Review of Agricultural Economics* 47, no. 3 (2020): 849–892.
- [61] A. Gümüşçü, M. E. Tenekeci, and A. V. Bilgili, "Estimation of Wheat Planting Date Using Machine Learning Algorithms Based on Available Climate Data," *Sustainable Computing: Informatics and Systems* 28 (2020): 100308.
- [62] D. Paudel, H. Boogaard, A. de Wit, et al., "Machine Learning for Large-Scale Crop Yield Forecasting," *Agricultural Systems* 187 (2021): 103016.
- [63] A. Wolanin, et al., "Estimating Crop Primary Productivity With Sentinel-2 and Landsat 8 Using Machine Learning Methods Trained With Radiative Transfer Simulations," *Remote Sensing of Environment* 225 (2019): 441–457.
- [64] E. A. Abioye, O. Hensel, T. J. Esau, et al., "Precision Irrigation Management Using Machine Learning and Digital Farming Solutions," *AgriEngineering* 4, no. 1 (2022): 70–103.
- [65] E. M. Dogo, N. I. Nwulu, B. Twala, and C. Aigbavboa, "A Survey of Machine Learning Methods Applied to Anomaly Detection on Drinking-Water Quality Data," *Urban Water Journal* 16, no. 3 (2019): 235–248.
- [66] M. Ilić, Z. Srdjević, and B. Srdjević, "Water Quality Prediction Based on Naïve Bayes Algorithm," *Water Science and Technology* 85, no. 4 (2022): 1027–1039.
- [67] J.-S. Chou, C.-C. Ho, and H.-S. Hoang, "Determining Quality of Water in Reservoir Using Machine Learning," *Ecological Informatics* 44 (2018): 57–75.
- [68] S. Khullar and N. Singh, "Machine Learning Techniques in River Water Quality Modelling: A Research Travelogue," *Water Supply* 21, no. 1 (2021): 1–13.
- [69] L. Nobrega, A. Tavares, A. Cardoso, and P. Goncalves, "Animal Monitoring Based on IoT Technologies," in *2018 IoT Vertical and Topical Summit on Agriculture - Tuscany (IOT Tuscany)*, (Tuscany, Italy: IEEE, 2018), 1–5.
- [70] M. Bonneau, J.-A. Vayssade, W. Troupe, and R. Arquet, "Outdoor Animal Tracking Combining Neural Network and Time-Lapse Cameras," *Computers and Electronics in Agriculture* 168 (2020): 105150.
- [71] R. García, J. Aguilar, M. Toro, A. Pinto, and P. Rodríguez, "A Systematic Literature Review on the Use of Machine Learning in Precision Livestock Farming," *Computers and Electronics in Agriculture* 179 (2020): 105826.
- [72] D. Zhu, C. Cai, T. Yang, and X. Zhou, "A Machine Learning Approach for Air Quality Prediction: Model Regularization and Optimization," *Big Data and Cognitive Computing* 2, no. 1 (2018): 5–15.
- [73] Z. Ghaemi, A. Alimohammadi, and M. Farnaghi, "LaSVM-Based Big Data Learning System for Dynamic Prediction of Air Pollution in Tehran," *Environmental Monitoring and Assessment* 190 (2018): 300.
- [74] M. Ren, G. Cheng, W. Zhu, W. Nie, K. Guan, and T. Yang, "A Prediction Model for Surface Deformation Caused by Underground Mining Based on Spatio-Temporal Associations," *Geomatics, Natural Hazards and Risk* 13, no. 1 (2022): 94–122.
- [75] N. Li, H. Nguyen, J. Rostami, W. Zhang, X.-N. Bui, and B. Pradhan, "Predicting Rock Displacement in Underground Mines Using Improved Machine Learning-Based Models," *Measurement* 188 (2022): 110552.